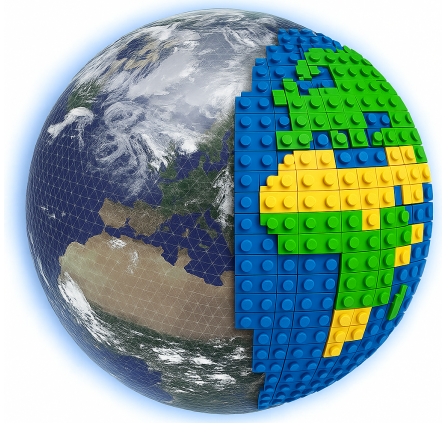




legoESM Scientific Guide

A Fully Differentiable Earth System Model in JAX

Version 0.1 — June 2026

legoESM Development Team

This document provides a comprehensive scientific description of the legoESM Earth System Model, including governing equations, numerical methods, and parameterisation schemes for all model components. The structure follows the NCAR CAM Scientific Guide convention of self-contained chapters for dynamics, physics, ocean, land, and sea ice.

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Chapter 1

Introduction

1.1 Overview

legoESM is a modular, fully differentiable Earth System Model (ESM) written entirely in JAX. Every component—atmosphere, ocean, land, sea ice, lake, and biogeochemistry—is compatible with JAX’s automatic differentiation (`jit`, `grad`, `vmap`, `scan`), enabling novel applications in data assimilation (4D-Var with exact adjoints), parameter estimation, hybrid AI–physics modelling, and sensitivity analysis.

The model features:

- **25+ dynamical cores** spanning five discretisation families (C-D grid FV3, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi), with shallow water, hydrostatic primitive equation, and non-hydrostatic compressible Euler equation sets;
- **35+ physics parameterisations** for radiation, convection, microphysics, boundary-layer turbulence, clouds, gravity wave drag, and surface exchange;
- A **3D Boussinesq ocean** with split-explicit barotropic solver, KPP vertical mixing, GM-Redi eddy parameterisation, and biogeochemistry (abiotic carbon cycle + NPZD ecosystem);
- A **multi-layer land model** with Richards-equation soil hydrology, Johansen thermal diffusion, Farquhar–Ball–Berry photosynthesis, and a DALEC-990 carbon cycle;
- **Sea-ice dynamics** with EVP rheology, multi-category thickness distribution, and thermodynamic coupling;
- A **tile-based coupler** with Monin–Obukhov surface layer, area-weighted flux blending, and radiation sub-cycling;
- **CMIP6-class infrastructure**: CF/CMOR output, six experiment templates, checkpointing with SHA-256 reproducibility verification, and a 16-parameter tuning registry;
- A **Spherical Fourier Neural Operator** (SFNO) for learned dynamics and physics emulation.

1.2 Design Principles

1. **JAX-first**: All code uses pure-functional JAX with no Python loops in hot paths. Conditional branching uses `jnp.where` or `jax.lax.cond` to preserve differentiability.
2. **Grid-agnostic physics**: Operators implement a shared protocol (`GridProtocol`); physics schemes accept any grid via column adapters that reshape to `(ncol, nlev)`.
3. **Modular composition**: Each parameterisation follows a factory pattern returning a callable

`physics_fn(state, grid, coord) → Tendencies.`

4. **Unified C-D grid:** Atmosphere and ocean share the same FV3-style C-D grid operator module on the cubed sphere, eliminating code duplication and the Hollingsworth–Kallberg instability.
5. **Pytree registration:** `Field` and state objects are registered as JAX pytrees for seamless autodiff through the entire model.

1.3 Notation and Conventions

Throughout this document:

- Vectors are denoted in bold: $\mathbf{v} = (u, v)$;
- $\hat{f}$ denotes spectral (spherical harmonic) coefficients;
- Subscript k indexes vertical levels (increasing downward);
- Half-integer subscripts ($k \pm \frac{1}{2}$) denote layer interfaces;
- SI units are used unless otherwise noted;
- $f = 2\Omega \sin \phi$ is the Coriolis parameter.

Chapter 2

Software Architecture and Package Federation

The scientific content of the following chapters maps onto a concrete software structure: legoESM is a *federation* of independently-installable Python packages that all share one import namespace. This chapter documents that structure — one subsection (“subchapter”) per package — so the equations of later chapters can be located in code.

2.1 Design: one namespace, independently-installable members

legoESM is a `uv` workspace of eight distribution packages laid out under `packages/<member>/legoesm/<subpkg>/`. All members ship into a single `legoesm` PEP-420 *namespace package*, so user code always writes `import legoesm.<subpkg>` regardless of which wheel provides it. The design goal is that each Earth-system component can be installed and run on its own—`pip install legoesm-ocean` yields a working ocean model—while a single meta-package `legoesm` pulls the whole stack.

The members obey a strict one-way **dependency DAG**, enforced in CI by `import-linter` contracts (`lint-imports`):

$$\underbrace{\text{core}}_{\text{substrate}} \prec \underbrace{\{\text{atmosphere, ocean, land, ice}\}}_{\text{components}} \prec \underbrace{\{\text{coupler, ml, tools}\}}_{\text{orchestration}}.$$

Components depend *only* on `legoesm-core` and never on one another or on the coupler; shared physics (e.g. the FV3 shallow-water barotropic solver reused by the ocean) is exposed through the substrate or via `importlib.metadata` entry points (`legoesm.sw_barotropics`), not direct cross-component imports.

2.2 The substrate: `legoesm-core`

`legoesm-core` ships the subpackages `core` (fields, operators, state, conservation accumulators, EOS), `grids` (cubed-sphere, lat-lon, Gaussian/spectral, MPAS/Voronoi, vertical coordinates, and the grid+complexity *capability matrix*; Chapter 3), `runtime` (device/precision/backend bootstrap), `parallel` (halo exchange and reductions; Chapter on parallelism), `io`, `timestepping` (RK integrators), and `components` (the complexity-ladder taxonomy and component protocols), together with the loose modules `constants`, `thermo`, `registry`, and `surface_albedo`. It imports nothing above itself.

2.3 Earth-system components

2.3.1 `legoesm-atmosphere`

The `atmosphere` subpackage: all atmospheric dynamical cores (Chapter 5)—cubed-sphere C-D grid, spectral, lat-lon finite-volume and C-grid, MPAS/Voronoi, and SFNO—the physics parameterisation packages (radiation, convection, microphysics, boundary layer, clouds, gravity-wave drag), and the dycore-free single-column model (SCM).

2.3.2 `legoesm-ocean`

The `ocean` subpackage: the 3-D split-explicit Boussinesq ocean (baroclinic/barotropic on cubed-sphere, lat-lon C-grid, and MPAS), vertical and lateral mixing (KPP, GM-Redi, tidal), bottom drag and surface forcing, biogeochemistry, the idealized-experiment suite, the centennial spin-up library, and the peer-comparison fidelity harness.

2.3.3 `legoesm-land`

The `land` subpackage: slab (bucket-hydrology) and multi-layer (Richards-equation, Johansen-thermal) land, the DALEC carbon cycle, snow, and PFT-weighted parameter providers.

2.3.4 `legoesm-ice`

The `ice` subpackage: thermodynamic slab and EVP/mEVP dynamic sea ice with the multi-category ice-thickness distribution and optional snow/brine/ ridging/pond physics.

2.4 Orchestration: coupler, ML, and tools

2.4.1 `legoesm-coupler`

The `coupler` subpackage (tile-based surface coupling, bulk-flux and surface-energy closure; Chapter 4) and the `driver` subpackage (`ModelDriver` for atmosphere-only AMIP, `CoupledESMDriver` for full coupling, `EarthSystemDriver` for arbitrary compositions, plus the reproducibility *run manifest*).

2.4.2 `legoesm-ml`

The `ml` (SFNO blocks, spectral convolutions, conservation-aware losses), `training` (differentiable training driver, ERA5 ingestion, S2S workflows), and `da` (variational 4D-Var on `jax.grad` through the compiled segment kernel) subpackages.

2.4.3 `legoesm-tools`

The `forcing` (GHG/ozone/aerosol/SST/runoff loaders), `diagnostics` (energy/mass budgets, column integrals, and the conservation PASS/FAIL *gates*), `experiments` (including the shared component-agnostic *test-matrix framework*, `experiments.matrix`), and `visualization` subpackages.

2.5 Reproducibility, scripts, and experiments

Runnable code lives in bucketed directories under `scripts/`: `run/` (production drivers), `matrix/` (the complexity-tiered test-matrix runners), `experiment/` (the template/provenance harness: `init_experiment`, `validate_templates`, `fetch_data`, `lego_detect_machine`), and `bench/`, `plot/`, `validate/`, `data/`. Experiments are versioned, tier-annotated YAML *templates* (`config/templates/`) resolved into a runnable directory; every run writes a `run_manifest.json` (resolved config, state digest, git commit, and library versions) so that `legoesm reproduce` can re-execute it bit-for-bit. Tests carry complexity-tier markers `tier0–tier3` (research $\rightarrow$ operational) and assert conservation through the same gates as the matrix runners.

Chapter 3

Grid Systems and Coordinates

3.1 Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection** (Ronchi et al., 1996; Putman and Lin, 2007). This quasi-uniform tiling avoids the polar singularity and convergence of meridians inherent to a latitude–longitude mesh, so the maximum stable timestep is set by a near-uniform grid spacing rather than by the vanishing zonal spacing at the poles. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be the equiangular gnomonic coordinates on a face; great circles map to straight lines in (α_x, α_y) , which is what makes the projection convenient for conservative finite-volume discretisation (Sadourny, 1972).

3.1.1 Face-to-Cartesian Mapping

For face 0 (+ x face):

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y). \quad (3.1)$$

Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, with geographic coordinates $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

3.1.2 Cell Area

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (3.2)$$

Here R is the planetary radius, $\Delta \alpha$ the uniform angular cell width for a C_n face of $n \times n$ cells, and the metric factor D is the distance from the cube centre to the projected point. The D^{-3} weighting concentrates area away from the face centres and is exactly the Jacobian of the gnomonic map; integrating ΔA over the six faces recovers $4\pi R^2$, which is the discrete area-conservation property the finite-volume operators rely on.

3.1.3 Grid Spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (3.3)$$

The minimum grid spacing on a C_n cubed sphere is

$$\Delta x_{\min} = \frac{\pi R}{2n\sqrt{3}}, \quad (3.4)$$

occurring at the cube corners where the gnomonic distortion is maximal.

3.1.4 Grid Angle and Wind Rotation

The angle θ between the grid x -axis and geographic east:

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (3.5)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (3.6)$$

3.1.5 Halo Exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back, because the grid-aligned basis is discontinuous across a panel edge and a naive copy of (u_g, v_g) would inject a spurious rotation at the seam. A discrete duo-grid treatment of the panel edges removes the residual edge imprint that a single-grid halo leaves on differential operators (Mouallem et al., 2023). Two halo widths are supported: `halo = 1` for second-order and MUSCL stencils, and `halo = 2` for PPM/WENO5 and biharmonic operators.

Native 4D halo exchange. All 3D operators use a 4D halo path (`pad_halo_4d`, `pad_halo_vector_4d`) that exchanges all vertical levels in a single MPI message per neighbour, replacing the earlier `vmap(pad_halo)` path which issued n_{lev} separate messages.

MPI differentiability. All `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp` in `halo_exchange.py`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Only `allreduce(SUM)` is AD-safe among MPI reductions; `MAX/MIN/allgather/bcast` are used only in diagnostics and are not differentiable.

3.2 Gaussian Grid and Spectral Transforms

The spectral dynamical cores represent fields by their spherical-harmonic coefficients and use a Gaussian grid with triangular truncation at total wavenumber $n_{\max}$ (Bourke, 1972; Hoskins and Simmons, 1975). Triangular truncation is chosen because it is isotropic: the retained set of harmonics is invariant under rotation of the sphere, so no direction is preferentially resolved.

3.2.1 Grid Dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\max} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\max} + 1)(n_{\max} + 2)}{2}. \quad (3.7)$$

Here n_{lat} , n_{lon} , and n_{sh} are the number of latitudes, longitudes, and retained spectral coefficients. The $\frac{3}{2}$ factor is the standard transform-grid choice that makes the Gaussian quadrature exact for the quadratic (advective) products, eliminating spectral aliasing. Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced so the zonal direction can be treated by FFT.

3.2.2 Associated Legendre Polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. Computed via the recursion:

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (3.8)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (3.9)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

3.2.3 Spherical Harmonic Transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (3.10)$$

The transform is split into a fast Fourier transform in longitude followed by a Gaussian-quadrature sum over latitude μ_k weighted by w_k and the Legendre function P_n^m (Bourke, 1972). Because the quadrature is exact at this resolution, analysis followed by synthesis returns the resolved field to round-off.

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\text{max}}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (3.11)$$

3.2.4 Spectral Operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

3.2.5 Velocity from Vorticity and Divergence

Streamfunction $\psi = \nabla^{-2}\zeta$, velocity potential $\chi = \nabla^{-2}\delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (3.12)$$

where θ is colatitude.

3.3 Lat-Lon Grid

A uniform $n_{\text{lat}} \times n_{\text{lon}}$ latitude-longitude grid with A-grid (collocated) staggering. All operators use second-order centred differences. A Fourier polar filter is applied to tendencies at high latitudes to prevent the CFL constraint from becoming prohibitively small near the poles.

3.4 Voronoi/MPAS Grid

An unstructured Voronoi tessellation generated from an icosahedral parent mesh (Skamarock et al., 2012). The TRiSK (Thuburn–Ringle–Skamarock–Klemp) scheme provides mimetic discrete operators on this C-grid (Thuburn et al., 2009; Ringle et al., 2010). Mimetic here means the discrete divergence, gradient, and curl satisfy the same vector-calculus identities as their continuous counterparts ($\nabla \cdot \nabla \times = 0$ and $\nabla \times \nabla = 0$ to machine precision), which in turn guarantees discrete conservation of mass and potential vorticity and suppresses the spurious computational modes that a non-mimetic C-grid would admit. The placement is

- Scalars (height, tracers, pressure) at cell centres;
- Normal velocity at edges;
- Vorticity at vertices (triangle dual).

Variable resolution is supported by mesh refinement.

3.5 Vertical Coordinates

3.5.1 Sigma Coordinate

$$\sigma = p / p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (3.13)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$.

Simmons–Burridge Geopotential

The vertical finite differencing follows Simmons and Burridge (1981), which is constructed so that the discrete pressure-gradient and energy-conversion terms conserve total energy under adiabatic, frictionless flow. The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (3.14)$$

where the Simmons–Burridge coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta \sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (3.15)$$

Here Φ_s is the surface geopotential, R_d the dry gas constant, $T_{k'}$ the layer temperature, and α_k the Simmons–Burridge weighting that distributes the within-layer geopotential increment consistently with the hydrostatic relation, so that the discrete hydrostatic balance is exact.

Sigma-Dot (Vertical Velocity)

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta \sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta \sigma_{k'}. \quad (3.16)$$

Pressure Velocity

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}. \quad (3.17)$$

3.5.2 Hybrid Sigma-Pressure Coordinate

$$p(k) = A(k) p_{\text{ref}} + B(k) p_s, \quad (3.18)$$

where $A(k)$ and $B(k)$ are the prescribed hybrid coefficients with $A(k) \rightarrow 1$, $B(k) \rightarrow 0$ at the model top (pure pressure levels) and $A(k) \rightarrow 0$, $B(k) \rightarrow 1$ near the surface (pure sigma). Flattening the upper levels to pressure surfaces eliminates the large pressure-gradient force errors over steep topography that afflict the pure sigma coordinate, where the two large terms of the pressure gradient must cancel along a steeply sloping surface.

3.5.3 Terrain-Following Height Coordinate (z^*)

Used by the non-hydrostatic compressible Euler equations:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (3.19)$$

A 1D reference state ($\rho_0(z)$, $\theta_0(z)$, $\pi_0(z)$) is pre-computed for the perturbation formulation.

3.5.4 Ocean z^* Coordinate

The ocean uses a dynamic z^* coordinate with free-surface elevation η :

$$J(\eta) = \frac{\eta + H}{H_{\text{max}}}, \quad (3.20)$$

where H is the resting depth and J the layer-thickness Jacobian. Stretching each column by $J(\eta)$ lets every level dilate uniformly with the moving free surface, so the coordinate stays a smooth function of the surface state, in the same spirit as the Lagrangian-then-remap vertical treatment of the finite-volume atmosphere core ([Lin, 2004](#)).

Chapter 4

Coupling of Dynamics and Physics

4.1 Physics Pipeline

The `PhysicsPipeline` class composes all physical parameterisations into a single callable that is invoked once per dynamics timestep. The ordering is not arbitrary: each parameterisation operates on the state already updated by the preceding ones, so radiation sets the column heating that drives convection, convection conditions the moisture field seen by the microphysics, and the boundary layer mixes the resulting profiles before the surface fluxes close the column energy and water budgets. The calling sequence within a single step is:

1. **Dynamics step:** Update winds, temperature, surface pressure, and tracers via the selected dynamical core.
2. **Physics step:** Apply parameterisations in the following order:
 - (a) Radiation (with optional sub-cycling),
 - (b) Deep convection,
 - (c) Shallow convection / large-scale condensation,
 - (d) Cloud microphysics,
 - (e) Boundary-layer turbulence and vertical diffusion,
 - (f) Gravity wave drag,
 - (g) Surface fluxes (tile coupler).
3. **Conservation fixers:** Mass, moisture, and optionally energy.
4. **Diagnostics:** Energy budget, monthly means, snapshots.

4.2 Radiation Sub-Cycling

Radiation is the most expensive physics component. The pipeline supports calling radiation every N_{rad} dynamics steps while holding radiative heating rates constant between calls. The branching is implemented via `jax.lax.cond` with both branches returning identical array shapes and dtypes.

4.3 Grid-Agnostic Column Adapter

The `ColumnAdapter` reshapes any grid's 3D state into a `(ncol, nlev)` column format for physics, then unflattens back to the native grid shape. This allows all physics schemes to operate on simple column arrays regardless of the underlying horizontal discretisation.

4.4 Time-Split vs Process-Split

legoESM uses **time-split** coupling: dynamics and physics operate sequentially within each timestep, so the physics sees the fully advected state rather than a partial tendency. Physics tendencies are computed from the dynamics-updated state and applied as a forward-Euler update. In process-split coupling, by contrast, dynamics and physics tendencies are evaluated from the same time-level state and summed; the sequential update chosen here keeps each parameterisation consistent with the column the previous one produced, and is natural for the finite-volume dycores that remap onto a Lagrangian vertical coordinate internally before physics is applied.

Chapter 5

Atmospheric Dynamics

This chapter describes the governing equations and numerical methods for each dynamical core. Cores are organised by equation system (shallow water, hydrostatic primitive equations, non-hydrostatic compressible Euler), then by discretisation.

5.1 Shallow Water Equations

The shallow water system is the canonical proving ground for horizontal dynamical cores: it retains the rotational and gravity-wave dynamics that control the fast modes of the full equations while collapsing the vertical structure to a single layer, so the standard test set of [Williamson et al. \(1992\)](#) can isolate grid imprinting, dispersion, and conservation errors. The rotating shallow water equations in **vector-invariant form**:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (5.1)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (5.2)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (5.3)$$

where h is the fluid depth, $\mathbf{v} = (u, v)$ the horizontal velocity, f the Coriolis parameter, g gravity, and h_s the bottom topography. $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ is the Bernoulli function, $\zeta = \partial v / \partial x - \partial u / \partial y$ is the relative vorticity, and D_u, D_v represent hyperdiffusion. The first equation expresses exact mass (volume) conservation in flux form; the momentum equations express the balance between the absolute-vorticity $(\zeta + f)$ flux, the Bernoulli gradient that combines kinetic energy and free-surface pressure, and scale-selective dissipation. Writing the advection through $(\zeta + f)$ rather than as $\mathbf{v} \cdot \nabla \mathbf{v}$ keeps the Coriolis and metric terms in a single discretely consistent operator, which is what permits simultaneous mass, energy, and potential-enzrophy control on the staggered grids below.

5.1.1 C-D Grid Cubed-Sphere Discretisation

This is the primary shallow water solver. It maps the equations onto the gnomonic cubed sphere ([Ronchi et al., 1996](#); [Putman and Lin, 2007](#)), a quasi-uniform tiling of six logically rectangular panels that avoids the polar singularity of the lat-lon mesh, and uses the FV3-style C-D grid staggering of [Lin \(2004\)](#) and [Putman and Lin \(2007\)](#). The C-D arrangement carries momentum on the D-grid (so vorticity is cleanly defined at corners) while diagnosing C-grid mass fluxes for

transport, combining the favourable geostrophic adjustment of the C-grid with the well-resolved vorticity dynamics of the D-grid.

Prognostic variables. D-grid winds (u_d, v_d) at cell corners, shape $(6, n+1, n+1)$; height h at cell centres, shape $(6, n, n)$.

D-grid vorticity. Vorticity at each cell corner is the discrete circulation divided by the corner dual-cell area:

$$\zeta_{\text{corner}} = \frac{1}{A_c} \oint \mathbf{u} \cdot d\mathbf{l} \approx \frac{1}{A_c} [u_{i,j}^D \Delta x - u_{i+1,j}^D \Delta x + v_{i,j+1}^D \Delta y - v_{i,j}^D \Delta y]. \quad (5.4)$$

Here A_c is the corner dual-cell area and $\Delta x, \Delta y$ are the local edge lengths. Because vorticity is defined as the line integral of the tangential wind around the dual cell, the discrete curl inherits the Stokes-theorem identity exactly, so the kinetic-energy and enstrophy budgets remain mutually consistent. This circulation-based discretisation is exact on the D-grid and **eliminates the Hollingsworth–Kallberg instability** that plagues A-grid vorticity operators, in which a vorticity stencil inconsistent with the kinetic-energy gradient feeds a spurious symmetric-instability mode.

Arakawa–Lamb gradient. The Bernoulli gradient at D-grid corners uses a 4-point average:

$$\left. \frac{\partial B}{\partial x} \right|_{i+\frac{1}{2}, j+\frac{1}{2}} = \frac{B_{i+1,j} + B_{i+1,j+1} - B_{i,j} - B_{i,j+1}}{2\Delta x}. \quad (5.5)$$

C-grid mass flux. C-grid velocities are diagnosed from D-grid winds by face-edge averaging. Scalar transport uses upwind C-grid mass fluxes:

$$\frac{dh}{dt} = -\frac{1}{A} [(hu_c)_{i+\frac{1}{2},j} - (hu_c)_{i-\frac{1}{2},j} + (hv_c)_{i,j+\frac{1}{2}} - (hv_c)_{i,j-\frac{1}{2}}]. \quad (5.6)$$

FV3-faithful PPM transport (v3.9). As of v3.9, the C-D grid operator module underwent a complete rewrite to implement FV3-faithful Piecewise Parabolic Method (PPM) transport (Colella and Woodward, 1984) on staggered D/C grids. Key elements include:

- PPM reconstruction with Colella–Woodward monotonicity limiters for all scalar fields;
- Flux-form mass transport on staggered D-grid and C-grid;
- Corrected geographic-to-grid wind rotation eliminating sign errors in cross-face interpolation;
- Tuned hyperdiffusion with physical e -folding time ($\nu_4 = \Delta x^4 / \tau_{\text{efold}}$).

The module spans approximately 740 lines of JAX code.

FV3 forward–backward shallow-water core (v4.0). An FV3-exact implementation of the complete `c_sw/d_sw` forward–backward stepping algorithm is provided in `fv3_sw_core.py`. The forward step (`c_sw`) computes C-grid mass fluxes and height tendencies; the backward step (`d_sw`) updates D-grid winds using the vorticity flux and Bernoulli gradient.

d2a2c_vect. The D-grid $\rightarrow$ A-grid $\rightarrow$ C-grid vector conversion faithfully ports the GFDL FV3 algorithm, including:

- Edge-boundary stencils with one-sided 4-point interpolation at cube-face edges;
- Non-orthogonality corrections via $\cos \alpha$ and $(\sin^2 \alpha)^{-1}$ corner metrics;
- Halo-aware padding with face-boundary detection.

PPM wind and vorticity transport. C-grid wind tendencies (`xtp_u`, `ytp_v`) and absolute vorticity flux use Piecewise Parabolic Method reconstruction with FV3-specific interpolation coefficients (A_1 , A_2 , C_1 – C_3) and Colella–Woodward monotonicity limiters.

Cell-centred kinetic energy. Kinetic energy is computed from A-grid velocities at cell centres, with an optional 4th-order averaging stencil (`a2b_ord4`).

Duo-grid sub-grid metrics (Mouallem et al. 2023). Following the duo-grid construction of Mouallem et al. (2023), the cubed-sphere C-D grid computes 9-point sub-grid angle metrics per cell:

$$\sin \alpha_{\text{sg}}, \cos \alpha_{\text{sg}} \in \mathbb{R}^{6 \times n \times n \times 9}, \quad (5.7)$$

encoding the local deviation from orthogonality at cell centres, edge midpoints, and corners. These enable FV3-faithful flux scaling at cube-face boundaries where the gnomonic projection introduces non-orthogonality. Additional C-D grid metrics:

- Centre-to-centre distances Δx_c (shape $6 \times (n+1) \times n$) and Δy_c (shape $6 \times n \times (n+1)$);
- Non-orthogonality factors $\cos \alpha_{\text{corner}}$ and $(\sin^2 \alpha_{\text{corner}})^{-1}$ at D-grid corners.

Fidelity calibration and 3D edge-artifact sentinels. The C-D-grid FV3 port is gated on a dual-target Williamson 2 / Williamson 5 calibration drawn from the standard sphere test set (Williamson et al., 1992): at C36 with $\Delta t = 300$ s the production knobs `div_damp_order = 8` and `damp_v = 0.030` (iter-1030) satisfy both criteria simultaneously — Williamson 2 v-wind error $|v_{\parallel}| < 0.119 \text{ m s}^{-1}$ and Williamson 5 day-5 height $h > 0$ with $|\mathbf{u}| < 80 \text{ m s}^{-1}$. A 48-test sentinel suite exercises the full 3D coverage matrix (hydrostatic / non-hydrostatic $\times$ rest / edge / long integration $\times$ cube-edge $i, j \in \{0, n-1\}$) and a long 100-step cosine-bell closure within $\pm 5\%$. Visual verification of v -wind snapshots is mandatory — error norms can improve while edge artifacts shift in character; see `docs/fv3_fortran_fidelity_review.md` and `docs/cubed_sphere_edge_artifacts.md` for the iter-1045 status snapshot.

5.1.2 Lat-Lon Finite Volume Discretisation

A-grid staggering on a uniform $n_{\text{lat}} \times n_{\text{lon}}$ grid. Centred differences for all operators. A Fourier polar filter damps meridional wavelengths near the poles to maintain CFL stability.

5.1.3 C-Grid Lat-Lon Discretisation

Height at cell centres; u at longitude interfaces; v at latitude interfaces. The operators follow the energy- and enstrophy-conserving finite-difference construction of Sadourny (1972) for the sphere. Advantages: exact pressure gradient (just cell differences), mass flux naturally at cell interfaces (no interpolation), and better gravity wave dispersion. The Coriolis term requires 4-point interpolation, which is the price the C-grid pays for placing u and v on different interfaces; the interpolation weights are chosen to preserve the discrete energy conversion between kinetic and potential reservoirs.

5.1.4 TRiSK Discretisation on Voronoi Mesh

The unstructured Spherical Centroidal Voronoi Tessellation places scalars at cell centres and the normal velocity component u_e on edges. The TRiSK (Thuburn–Ringler–Skamarock–Klemp) family of mimetic operators (Thuburn et al., 2009; Ringler et al., 2010) is built so that the discrete

divergence, curl, and gradient satisfy the same vector-calculus identities as the continuum ($\nabla \cdot \nabla \times = 0$, $\nabla \times \nabla = 0$), guaranteeing exact mass conservation and a consistent potential-vorticity budget on a fully unstructured mesh. Following Ringler et al. (2010), the discrete operators on the Voronoi mesh are:

$$\text{Divergence at cell } c : (\nabla \cdot \mathbf{u})_c = \frac{1}{A_c} \sum_e n_{e,c} u_e l_e, \quad (5.8)$$

$$\text{Gradient at edge } e : (\nabla \phi)_e = \frac{\phi_{c2} - \phi_{c1}}{d_e}, \quad (5.9)$$

$$\text{Curl at vertex } v : (\nabla \times \mathbf{u})_v = \frac{1}{A_v} \sum_e t_{e,v} u_e d_e, \quad (5.10)$$

where $n_{e,c}$ and $t_{e,v}$ are orientation signs, l_e is edge length, d_e is distance between adjacent cell centres, and A_c, A_v are cell/vertex areas.

5.1.5 Spectral Shallow Water

The state is represented by spherical-harmonic coefficients and advanced in the vorticity–divergence formulation of Bourke (1972), in which the prognostic scalars $\hat{\zeta}$ and $\hat{\delta}$ replace the vector momentum and the horizontal Laplacian is diagonal in spectral space:

$$\frac{\partial \hat{\zeta}}{\partial t} = -\nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}], \quad (5.11)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (E + \widehat{\Phi} + \Phi_s), \quad (5.12)$$

$$\frac{\partial \hat{\Phi}}{\partial t} = -\nabla \cdot (\widehat{\Phi} \mathbf{v}), \quad (5.13)$$

where $E = \frac{1}{2}(u^2 + v^2)$ is the kinetic energy, $\Phi = gh$ the free-surface geopotential, Φ_s the surface geopotential, and the hat denotes the forward spherical-harmonic transform. Pseudospectral: nonlinear products are computed on the grid, where they are local and cheap, then transformed back to spectral space, where the linear gravity-wave and diffusion terms are exact and free of grid imprinting.

5.2 Hydrostatic Primitive Equations

Prognostic variables: u, v (horizontal winds), T (temperature), p_s (surface pressure), and tracers q_i .

5.2.1 Continuous Equations

In σ -coordinates:

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (5.14)$$

$$\frac{\partial u}{\partial t} = (\zeta + f)v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (5.15)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f)u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (5.16)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (5.17)$$

Here $\sigma = p/p_s$ is the terrain-following vertical coordinate, $\dot{\sigma}$ the associated vertical velocity, $\omega = dp/dt$ the pressure velocity, Φ the geopotential, R_d the dry gas constant, and Q the diabatic heating from physics; $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, and $\kappa = R_d/c_p$. The surface-pressure tendency (5.14) is the vertically integrated mass continuity that closes the hydrostatic system: with the vertical momentum equation replaced by hydrostatic balance, mass divergence in the column—rather than vertical acceleration—sets the surface pressure. The pressure-gradient force splits into the along- σ Bernoulli gradient and the $R_d T \nabla \ln p_s$ term that accounts for the tilt of the σ surfaces. The adiabatic heating term $\kappa T \omega/p$ represents compression warming during descent and is the thermodynamic counterpart of the work done by the pressure-gradient force.

5.2.2 C-D Grid Cubed-Sphere Discretisation

Prognostic state: u_d, v_d at D-grid corners ($6, n+1, n+1, N_{\text{lev}}$); T and p_s at cell centres. Spatial operators are identical to the shallow water C-D grid (Section 5.1), extended to 3D with:

- Simmons–Burridge energy-conserving geopotential integration (Simmons and Burridge, 1981) (Section 3.5);
- Sigma-dot computation with discrete continuity closure;
- Upwind vertical advection: $-\dot{\sigma} \partial f / \partial \sigma$.

5.2.3 Spectral Transform Discretisation

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972):

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] + \widehat{\text{vert. adv.}} - \nu \nabla^2 \hat{\zeta}, \quad (5.18)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (K + \Phi + \widehat{R_d T \ln p_s}) + \widehat{\text{vert. adv.}} - \nu \nabla^2 \hat{\delta}, \quad (5.19)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T \mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \widehat{\kappa T \omega / p}, \quad (5.20)$$

$$\frac{\partial \ln p_s}{\partial t} = - \int_0^1 \hat{\delta} d\sigma. \quad (5.21)$$

Spectral curl and divergence. For flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$:

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (5.22)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices. The $\cos\phi$ weighting is absorbed into the Legendre basis, making the formulation pole-safe.

Semi-implicit treatment of gravity waves. Following Hoskins and Simmons (1975), the fast gravity-wave terms are treated implicitly. Linearising about a reference temperature $T_{\text{ref}} = 300$ K yields a Helmholtz equation per spectral mode:

$$[\mathbf{I} + \alpha^2 \Delta t^2 \lambda_n \mathbf{\Gamma}] \hat{\delta}^{n+1} = \hat{\delta}^{\text{explicit}}, \quad (5.23)$$

where $\lambda_n = n(n+1)/a^2$ is the Laplacian eigenvalue and $\mathbf{\Gamma}$ is the vertical coupling matrix built from the energy-conserving vertical finite differences of Simmons and Burridge (1981). Because the harmonics diagonalise the horizontal Laplacian, the implicit system decouples by wavenumber into a small block-tridiagonal solve in the vertical; this is pre-factored (LU decomposition) for each wavenumber n at initialisation, removing the gravity-wave CFL constraint and allowing 3–4 $\times$ larger timesteps.

Verification (v3.9). A comprehensive spectral dycore test suite validates conservation (mass, energy, enstrophy), long-integration stability under Held–Suarez forcing, spectral convergence with increasing truncation wavenumber, and semi-implicit scheme correctness. See Chapter 18 for details.

5.2.4 Lat-Lon Finite-Volume Discretisation

Identical equations to the C-D grid version, discretised on the A-grid lat-lon mesh with centred differences. Polar filter applied to all tendencies. Optional Rayleigh sponge layer above a configurable σ cutoff.

5.2.5 MPAS/Voronoi Discretisation

TRiSK operators (Section 5.1) extended to 3D, following the MPAS atmosphere formulation of Skamarock et al. (2012). Uses RK4 or SSP-RK3 (Shu and Osher, 1988) time stepping with energy- or enstrophy-conserving potential vorticity flux options; the choice of PV flux selects which quadratic invariant the discretisation conserves to round-off, trading energetic fidelity against the enstrophy control that suppresses spurious grid-scale vorticity.

5.3 Non-Hydrostatic Compressible Euler Equations

Dropping the hydrostatic approximation restores vertically propagating acoustic and gravity waves, so the core integrates the fully compressible Euler equations with a split-explicit, height-based formulation in the manner of the WRF/MPAS nonhydrostatic solvers (Klemp et al., 2007; Skamarock and Klemp, 2008). In the z^* terrain-following coordinate, the prognostic variables are perturbations from a hydrostatically balanced reference state $(\rho_0, \theta_0, \pi_0)$, which removes the dominant hydrostatic balance from the pressure-gradient and buoyancy terms and leaves only the dynamically active residuals:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (5.24)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (5.25)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (5.26)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (5.27)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (5.28)$$

Here w is the vertical velocity, θ' and ρ' the potential-temperature and density perturbations, π' the Exner-pressure perturbation, J the Jacobian of the terrain-following map, K the horizontal kinetic energy, and $r(z)$ the Rayleigh sponge coefficient. The horizontal momentum equations carry the non-hydrostatic pressure gradient $c_p \theta \nabla \pi'$ in place of the geopotential gradient, the w equation balances the same vertical pressure-gradient force against buoyancy $-g \theta' / \theta_0$, and the density equation is the unapproximated mass continuity. The Exner perturbation closes the system through the equation of state:

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (5.29)$$

5.3.1 Split-Explicit Time Integration

Acoustic modes propagate roughly an order of magnitude faster than the meteorologically relevant advective and gravity-wave modes, so an explicit single-rate scheme would be acoustically CFL-limited and wasteful. The core therefore time-splits the integration (Wicker and Skamarock, 2002; Klemp et al., 2007): the slow, meteorologically important terms are advanced on the long RK3 step, while the acoustically active terms are sub-cycled on short forward-backward substeps.

Slow modes (SSP-RK3): advection, Coriolis, horizontal pressure gradient force, and diffusion.

Fast modes (forward-backward acoustic substeps): vertical sound/gravity waves. Typically $N_s = 6$ substeps per RK stage. The forward-backward ordering—updating w from the current pressure, then the density and temperature from the just-updated w —makes the vertical acoustic substep implicitly stable without a tridiagonal solve.

Forward step (vertical velocity).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (5.30)$$

Backward step (density, potential temperature).

$$\rho'^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta'^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (5.31)$$

Acoustic off-centering. To damp vertically propagating acoustic/gravity wave noise (Skamarock and Klemp, 2008):

$$\rho_\beta^{m+1} = (1 + \beta) \rho'^{m+1} - \beta \rho'^m, \quad (5.32)$$

where $\beta \geq 0$ is the off-centering parameter (default 0.1).

5.3.2 Sponge Layer

$$r(z) = r_{\max} \sin^2\left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}}\right). \quad (5.33)$$

5.3.3 Turbulence Closure: CRM, LES, and DNS

The diffusion terms D_u, D_v, D_w, D_θ in Eqs. (5.24)–(5.27) are evaluated in flux form, $D_\phi = \rho_0^{-1} \partial_j (\rho_0 K_\phi \partial_j \phi)$ over the horizontal directions $j \in \{x, y\}$ and, optionally, the vertical $j = z^*$ (a mass-weighted leg matching the SAM `diffuse_*` operators). The *same* discrete operator serves a hierarchy of three turbulence-resolving regimes that differ only in the viscosity K_m and the grid spacing Δx ; the heat and scalar diffusivity is $K_h = K_m/\text{Pr}$.

CRM (convection-permitting), $\Delta x \sim 1\text{--}4\text{ km}$. The Smagorinsky–Lilly eddy viscosity (Smagorinsky, 1963)

$$K_m = (C_s \Delta)^2 \sqrt{\max(0, |S|^2 - \text{Pr} N^2)}, \quad \Delta = (\Delta x \Delta y \Delta z)^{1/3}, \quad (5.34)$$

with the full strain magnitude $|S|^2 = 2 S_{ij} S_{ij}$ (all six components, including the vertical-shear terms S_{13}, S_{23}, S_{33}), $C_s \approx 0.19$, and the moist sub-grid stratification cutoff N^2 (the SAM `dosmagor` correction) that suppresses mixing in statically stable layers. The energy-containing eddies are entirely sub-grid, so Eq. (5.34) parameterizes *all* turbulent transport. This is the configuration validated against gSAM for the GATE, LBA, and RCE deep convection cases.

LES (large-eddy simulation), $\Delta x \sim 10\text{--}100\text{ m}$. The *same* closure (5.34) with $C_s \approx 0.15$ and an optional near-wall mixing-length limiter $\ell_m = \min(C_s \Delta, \kappa z)$. The energy-containing eddies are now resolved and K_m models only the inertial-subrange stress. The gSAM reference simulations of GATE and LBA are themselves $\Delta x = 100\text{ m}$ LES, so the LES mode applies the identical closure at the identical resolution as the oracle.

DNS (direct numerical simulation), $\Delta x \rightarrow$ the Kolmogorov scale. No turbulence model. K_m is replaced by the constant molecular kinematic viscosity ν (with $K_h = \nu/\text{Pr}$ and $\text{Pr} \approx 0.71$ for air), yielding the molecular dissipation $\nu \nabla^2 \mathbf{u}$ directly; all turbulent scales down to dissipation are resolved.

The regime is selected by the `turbulence_closure` parameter (`smagorinsky` / `molecular` / `none`). Because the three regimes share the single discrete flux-form operator, LES and DNS are a configuration change rather than a separate code path. The molecular closure has been verified to reproduce the analytical viscous decay $\partial_t \hat{u} = -\nu k^2 \hat{u}$ of an isolated shear mode to within the time-integration error.

5.4 Finite-Volume Transport

Scalar advection in the finite-volume cores is cast in flux form, so the cell-averaged tracer changes only through fluxes across its faces; this enforces discrete conservation by construction. The remaining task is to reconstruct the sub-grid profile used to evaluate those face fluxes, trading formal order of accuracy against the monotonicity needed to keep tracers (mixing ratios, densities) physically bounded. The cores offer a hierarchy of reconstructions of increasing order.

5.4.1 MUSCL Reconstruction (2nd Order)

Piecewise-linear reconstruction with slope limiting (van Leer, 1979):

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (5.35)$$

with minmod or MC (monotonised central) limiters. The limiter ϕ reduces the cell slope wherever it would create a new extremum, yielding a second-order, total-variation-diminishing scheme that suppresses the spurious oscillations of unlimited centred differences near sharp gradients.

5.4.2 PPM Reconstruction (4th Order)

The Piecewise Parabolic Method (Colella and Woodward, 1984) constructs a monotone parabola in each cell. Fourth-order edge values:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (5.36)$$

Monotonicity constraint: $\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1})$. Colella–Woodward limiting prevents new extrema in the parabolic profile. The parabolic in-cell profile delivers fourth-order accuracy in smooth flow while the edge-value clipping and the flattening of the parabola near discontinuities preserve monotonicity, making PPM the default tracer scheme where both accuracy and positivity matter.

5.4.3 WENO5 Reconstruction (5th Order)

Three sub-stencil polynomials combined with nonlinear weights adapting to local smoothness (Jiang and Shu, 1996):

$$q_L = \sum_k \omega_k q^{(k)}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad \alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad (5.37)$$

where d_k are ideal weights (1/10, 6/10, 3/10) and β_k are smoothness indicators. The nonlinear weights ω_k collapse onto the optimal fifth-order stencil in smooth regions but smoothly drop any sub-stencil that straddles a discontinuity, giving high order without the Gibbs oscillations of a fixed wide stencil; ε is a small floor preventing division by zero where the field is locally constant.

5.4.4 Rusanov Flux

The reconstructed left and right interface states (q_L, q_R) are combined into a single-valued face flux by a local Lax–Friedrichs (Rusanov) approximate Riemann solver:

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (5.38)$$

where $c = \sqrt{g h}$ is the gravity-wave speed. The first term is the centred average and the second is an upwind dissipation proportional to the maximum local signal speed α , which provides exactly enough numerical viscosity to stabilise the discontinuity while remaining conservative.

5.5 Time Integration

5.5.1 SSP-RK3

The three-stage strong-stability-preserving (SSP) Runge–Kutta method (Shu and Osher, 1988) is the workhorse explicit integrator. Each stage is a convex combination of forward-Euler updates, so any monotonicity or positivity property that holds for a single forward-Euler step under the stage CFL also holds for the full higher-order step—exactly the guarantee the limited finite-volume reconstructions above need:

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (5.39)$$

$$\mathbf{u}^{(2)} = \frac{3}{4}\mathbf{u}^n + \frac{1}{4}\mathbf{u}^{(1)} + \frac{1}{4}\Delta t F(\mathbf{u}^{(1)}), \quad (5.40)$$

$$\mathbf{u}^{n+1} = \frac{1}{3}\mathbf{u}^n + \frac{2}{3}\mathbf{u}^{(2)} + \frac{2}{3}\Delta t F(\mathbf{u}^{(2)}). \quad (5.41)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

5.5.2 Robert–Asselin–Williams (RAW) Time Filter

The leapfrog scheme used by the semi-implicit spectral PE solver supports two time levels and therefore admits a spurious computational mode that must be damped. The classical Robert–Asselin filter (Asselin, 1972) suppresses this mode but at the cost of degrading the scheme to first-order accuracy and spuriously damping the physical mode. The Williams (2009) modification distributes the filter increment across the time levels to recover second-order accuracy:

$$d^n = \frac{\gamma}{2}(X^{n-1} - 2X^n + X^{n+1}), \quad (5.42)$$

$$X_{\text{filt}}^n = X^n + (1 - \alpha) d^n, \quad (5.43)$$

$$X_{\text{filt}}^{n+1} = X^{n+1} + \alpha d^n, \quad (5.44)$$

with $\alpha = 0.5$ (default). This restores second-order accuracy while preserving the damping of the computational mode.

5.5.3 CFL Conditions and Adaptive Hyperdiffusion

For an explicit core the timestep is bounded so that the fastest signal crosses at most a fraction C_{CFL} of the smallest grid cell per step:

$$\Delta t_{\text{max}} = \frac{C_{\text{CFL}} \Delta x_{\text{min}}}{(u_{\text{max}} + c_{\text{gw}}) \sqrt{n_{\text{dim}}}}, \quad (5.45)$$

where $c_{\text{gw}} = \sqrt{gH}$ is the gravity wave speed, u_{max} the peak advective speed, and the $\sqrt{n_{\text{dim}}}$ factor accounts for simultaneous propagation along each dimension on a multidimensional grid.

The maximum stable hyperdiffusion coefficient for p -th order:

$$\nu_p = S \cdot \frac{C(p) \Delta x^{2p}}{\Delta t}, \quad C(2) = \frac{1}{2}, \quad C(4) = \frac{1}{8}, \quad C(6) = \frac{1}{48}, \quad (5.46)$$

where $S \in (0, 1]$ is a safety factor. The physical e-folding time for $2\Delta x$ waves is $\tau_{\text{efold}} = \Delta x^{2p} / \nu_p$.

Chapter 6

Atmospheric Physics

6.1 Thermodynamic Preliminaries

6.1.1 Saturation Mixing Ratio

The saturation vapour pressure e_s sets the moisture capacity of the air and underlies condensation, evaporation, and convective triggering. It is obtained from the empirical Tetens (August–Roche–Magnus) fit, which approximates the Clausius–Clapeyron integral over liquid water:

$$e_s(T) = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d/R_v \approx 0.622, \quad (6.1)$$

where $T_c = T - 273.15$ K is the Celsius temperature, p the total pressure, and ε the ratio of dry-air to vapour gas constants that converts the partial-pressure ratio into a saturation mixing ratio q_s . Over ice the latent heat of sublimation L_s is larger, so the saturation curve is steeper; we evaluate it directly from the Clausius–Clapeyron form referenced to the freezing point T_{freeze} :

$$e_{s,i}(T) = 611.2 \exp\left[\frac{L_s}{R_v} \left(\frac{1}{T_{\text{freeze}}} - \frac{1}{T}\right)\right]. \quad (6.2)$$

6.1.2 Virtual Temperature

The buoyancy of moist, condensate-laden air is governed by the virtual temperature T_v , which is the temperature dry air would need to match the density of the moist parcel. Water vapour lightens the parcel (positive $0.608 q_v$) while suspended condensate loads it (negative q_c, q_r, q_i):

$$T_v = T (1 + 0.608 q_v - q_c - q_r - q_i). \quad (6.3)$$

6.1.3 Moist Adiabatic Lapse Rate

When a saturated parcel rises, condensation releases latent heat and damps the cooling relative to the dry-adiabatic rate. The moist lapse rate Γ_m balances adiabatic expansion against this latent-heat feedback, the correction terms in numerator and denominator scaling with $L_v q_s$:

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}. \quad (6.4)$$

6.2 Radiation

6.2.1 Gray Two-Stream Radiation

The gray scheme replaces spectrally-resolved transfer with a single broadband longwave channel whose optical depth is prescribed analytically, following the aquaplanet formulation of [Frierson et al. \(2006\)](#). This yields a smooth, inexpensive, and fully differentiable radiative forcing suitable for idealised and process-study configurations.

Longwave Optical Depth

The total longwave optical depth τ is the integral of absorber along the path. A prescribed background term, scaled to be larger in the tropics (ϕ latitude) and to increase toward the surface in σ , is augmented by an interactive water-vapour contribution proportional to the column vapour mass:

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \sum_k \tau_{\text{moist}} q_{v,k} \frac{\Delta p_k}{g}, \quad (6.5)$$

where the moisture optical depth per layer is $\Delta \tau_{\text{moist},k} = \tau_{\text{moist}} q_{v,k} \Delta p_k / g$, with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 0.0115 \text{ m}^2 \text{ kg}^{-1}$.

Two-Stream Longwave Fluxes

Each layer absorbs and re-emits in the gray band. Its transmittance $\mathcal{T}_k$ follows Beer's law in the layer optical depth $\Delta \tau_k$, and by Kirchhoff's law the emissivity ε_k is its complement:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (6.6)$$

The diffusivity factor $D \approx 5/3$ accounts for the hemispheric-mean angular integration, converting the slant-path absorption into an effective vertical two-stream transfer.

The hemispheric fluxes are then accumulated by two sweeps that propagate boundary emission through the absorbing/emitting column. The upward stream starts from surface emission ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$, with σ_{SB} the Stefan–Boltzmann constant and T_s the surface temperature) and gains each layer's Planck source:

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (6.7)$$

The downward stream starts from a dark top of atmosphere ($F_0^\downarrow = 0$) and accumulates emission on the way down:

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (6.8)$$

Shortwave (Beer–Lambert)

Solar heating is treated as a single attenuated beam. The top-of-atmosphere flux $S_0/4$ (the diurnal- and geometry-averaged solar constant) is absorbed exponentially with depth, and a fixed surface albedo α_s reflects the remainder back to space:

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-\tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}), \quad (6.9)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1361 \text{ W m}^{-2}$, $\alpha_s = 0.31$. Only the downward beam is absorbed; reflected SW escapes to space.

Heating Rate

The radiative temperature tendency follows from energy conservation: the heating of a layer equals the convergence of the net radiative flux $F_{\text{net}}^{\uparrow}$ across its interfaces, divided by its heat capacity per unit area $c_p \Delta p/g$:

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^{\uparrow} - F_{\text{net},k-1/2}^{\uparrow}}{p_{k+1/2} - p_{k-1/2}}. \quad (6.10)$$

Writing the heating as a flux difference makes it conservative by construction: summing over the column telescopes to the net top-of-atmosphere minus surface flux.

6.2.2 RRTMGP Correlated- k Radiation

For realistic radiative transfer the model offers a correlated- k backend that resolves the spectral dependence of gaseous absorption. It descends from the RRTM longwave model of [Mlawer et al. \(1997\)](#) and its modern, accelerator-friendly RRTMGP reformulation ([Pincus et al., 2019](#)), in which line-by-line absorption is reorganised into a small set of pseudo-monochromatic g -points per band. The RRTMGP backend uses JAX-RRTMGP with:

- 128 LW and 112 SW g -points for correlated- k gas absorption;
- Two-stream delta-Eddington radiative transfer equation;
- Spectral line absorption for H₂O, CO₂, CH₄, N₂O, O₃;
- Cloud optics via liquid/ice water path and effective radii lookup tables;
- Optional aerosol optical depth with prescribed single-scattering albedo ($\omega_0 = 0.93$) and asymmetry factor ($g = 0.70$).

Default trace-gas concentrations: CO₂=415 ppmv, CH₄=1900 ppbv, N₂O=332 ppbv.

6.2.3 Solar Geometry

The cosine of the solar zenith angle:

$$\mu_0 = \sin \phi \sin \delta + \cos \phi \cos \delta \cos h, \quad (6.11)$$

where δ is the solar declination and h is the hour angle. Daily-mean insolation and perpetual-equinox modes are also available.

6.3 Deep Convection

Five convection schemes are available, all fully JAX-differentiable.

6.3.1 Simplified Betts–Miller (SBM) Relaxation

The SBM scheme is an adjustment closure: rather than resolving updraughts, it relaxes temperature and moisture toward a post-convective reference profile. It follows the convective-adjustment philosophy of [Betts and Miller \(1986\)](#) in the simplified, single-timescale form of [Frierson \(2007\)](#).

CAPE Computation

The available energy for convection is the convective available potential energy (CAPE), the vertically integrated buoyancy of an ascending parcel relative to its environment. Only positively

buoyant layers ($T_{\text{parcel}} > T_{\text{env}}$) contribute:

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (6.12)$$

Smooth Trigger

Convection is switched on by a trigger function $\mathcal{F}$ that activates once CAPE exceeds a threshold CAPE_c . A sigmoid σ of sharpness κ_s replaces the discontinuous on/off switch, keeping the parameterisation differentiable for end-to-end training:

$$\mathcal{F} = \sigma(\kappa_s(\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (6.13)$$

Enthalpy-Conserving Relaxation

The reference profile T_{ref} is adjusted so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p(T_{\text{ref},k} - T_k) + L_v(q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (6.14)$$

This constraint guarantees that the latent heat released by drying the column exactly matches the sensible heating, so the adjustment redistributes energy without spuriously creating or destroying it. The state is then relaxed toward the reference on a fixed convective timescale τ_c , gated by the trigger $\mathcal{F}$:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (6.15)$$

Precipitation

Because the column dries during adjustment, surface precipitation is just the mass of vapour removed per unit area, recovered by integrating the moisture tendency over the column:

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (6.16)$$

6.3.2 Kuo Column Moisture-Excess Scheme

Convection proportional to column-integrated moisture excess above saturation [kg m^{-2}]. Heating and moistening are partitioned via a parameter α_{heat} :

$$MC = \int \max(q_v - q_s, 0) \frac{dp}{g}, \quad (6.17)$$

$$\frac{dT}{dt} = \mathcal{F} \alpha_{\text{heat}} \frac{T_{\text{moist}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} (1 - \alpha_{\text{heat}}) \frac{q_{s,\text{moist}} - q_v}{\tau_c}, \quad (6.18)$$

with $\alpha_{\text{heat}} = 0.75$ and $\tau_c = 3600 \text{ s}$.

6.3.3 Mass-Flux Convection (Arakawa–Wu Type)

In contrast to adjustment schemes, the mass-flux approach explicitly carries a cloud-base convective mass flux and an entraining updraught, in the spirit of the bulk mass-flux closures of [Tiedtke \(1989\)](#) and [Zhang and McFarlane \(1995\)](#). A prognostic base mass flux M_c relaxes toward an equilibrium value set by CAPE:

$$M_{\text{eq}} = \sigma[(\text{CAPE} - \text{CAPE}_{\text{thresh}})/\text{scale}] \cdot M_{\text{scale}}, \quad \frac{dM_c}{dt} = \frac{M_{\text{eq}} - M_c}{\tau_{\text{adj}}}. \quad (6.19)$$

The prognostic relaxation on timescale τ_{adj} lets the cloud ensemble respond to large-scale destabilisation with a finite memory rather than instantaneously equilibrating CAPE.

The mass-flux profile is sinusoidal: $m(p) = \sin[\pi(p_{\text{base}} - p)/\Delta p]$. The entraining updraft uses exponential dilution: as the plume rises it mixes in environmental air at rate ϵ_0 , so its properties interpolate between the undilute moist-adiabatic value and the environment:

$$T_u(z) = e^{-\epsilon_0 z} T_{\text{moist}} + (1 - e^{-\epsilon_0 z}) T_{\text{env}}, \quad (6.20)$$

with entrainment rate $\epsilon_0 = 10^{-3} \text{ m}^{-1}$ and detrainment rate $\delta_0 = 10^{-3} \text{ m}^{-1}$.

Precipitation. Precipitation is diagnosed from detrainment of updraft condensate, integrated over the column using layer thickness Δz :

$$P = \sum_k \delta_0 M_k q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{ s}^{-1}], \quad (6.21)$$

where $q_{\ell,k} = \max(q_{u,k} - q_{s,k}, 0)$ is the updraft condensate and $M_k = M_c m(p_k)$ is the mass flux profile $[\text{kg m}^{-2} \text{ s}^{-1}]$. The vertical measure Δz is used (not $\Delta p/g$) because M already carries units of mass flux per unit area, so $\delta_0 M q_{\ell}$ has units $[\text{kg m}^{-3} \text{ s}^{-1}]$ and multiplying by Δz [m] yields $[\text{kg m}^{-2} \text{ s}^{-1}]$.

6.3.4 EDMF Convection

Eddy-Diffusivity Mass-Flux with a prognostic updraft area fraction a_u :

$$\frac{da_u}{dt} = \frac{a_{u,\text{eq}} - a_u}{\tau_a}, \quad (6.22)$$

combining mass-flux (updraft) and eddy-diffusivity (environmental mixing) contributions.

Precipitation. As in the mass-flux scheme (§6.4), precipitation is diagnosed from condensate detrainment using Δz :

$$P = \sum_k \delta_0 M_{u,k} q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{ s}^{-1}], \quad (6.23)$$

with $M_{u,k} = \rho_k a_u w_{u,k}$ and $\delta_0 = 2 \times 10^{-3} \text{ m}^{-1}$.

6.3.5 Deep Convective Adjustment

The simplest parameterisation: scans bottom-to-top adjusting unstable layer pairs toward moist-adiabatic neutrality while conserving total enthalpy. Uses sigmoid activation for smooth, differentiable blending.

6.4 Cloud Microphysics

6.4.1 Kessler Warm-Rain Scheme

The single-moment Kessler scheme (Kessler, 1969) is the canonical warm-rain bulk parameterisation, tracking only condensate mass through the sequence saturation adjustment $\rightarrow$ autoconversion $\rightarrow$ accretion $\rightarrow$ evaporation $\rightarrow$ sedimentation. Prognostic variables: q_c (cloud water), q_r (rain).

Saturation adjustment. Supersaturation is removed by converting the excess vapour above q_s into cloud water. The saturation excess is converted to a tendency (rate) by dividing by the time step Δt :

$$\left. \frac{dq_v}{dt} \right|_{\text{sat}} = - \frac{\sigma(\kappa_s(q_v - q_s)) \cdot (q_v - q_s)}{\Delta t}, \quad (6.24)$$

with sigmoid sharpness κ_s . This ensures all microphysics outputs have consistent units of $\text{kg kg}^{-1} \text{s}^{-1}$.

Autoconversion (cloud $\rightarrow$ rain). Once cloud water exceeds a threshold $q_{c,\text{thresh}}$, collision and coalescence initiate rain at rate k_{auto} ; the `softplus` smooths the Kessler threshold so the onset stays differentiable:

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{auto}} \text{softplus}(q_c - q_{c,\text{thresh}}), \quad k_{\text{auto}} = 10^{-3} \text{ s}^{-1}, \quad q_{c,\text{thresh}} = 10^{-3} \text{ kg/kg}. \quad (6.25)$$

Accretion (collection of cloud by rain). Existing raindrops sweep up cloud droplets at a rate set by the product of cloud and rain content, the fractional power on q_r reflecting the mass-weighted fall-speed/cross-section of the rain spectrum:

$$\left. \frac{dq_c}{dt} \right|_{\text{accr}} = -k_{\text{accr}} q_c q_r^{0.875}, \quad k_{\text{accr}} = 2.2. \quad (6.26)$$

Rain evaporation.

$$\left. \frac{dq_r}{dt} \right|_{\text{evap}} = k_{\text{evap}} \frac{q_s - q_v}{q_s} q_r^{0.525}. \quad (6.27)$$

Sedimentation. Fall speed: $V_t = V_0 \sqrt{\rho_{\text{sfc}}/\rho}$ with $V_0 = 5 \text{ m/s}$.

Latent heating. $dT/dt = (L_v/c_p) \cdot (\text{condensation} - \text{evaporation})$.

6.4.2 Seifert–Beheng Two-Moment Warm Rain

The two-moment scheme of Seifert and Beheng (2001) predicts both mass and number, allowing the mean droplet size, and hence autoconversion efficiency, to evolve with the drop spectrum rather than being fixed. Prognostic mass (q_c, q_r) and number concentration (N_c, N_r).

Autoconversion. Autoconversion is keyed to the mean droplet mass $x_c = q_c \rho / N_c$: rain forms efficiently only once droplets grow past the separation mass x^* , the sigmoid σ providing a smooth onset in place of a hard size cutoff:

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{au}} q_c^2 \sigma[\kappa(x_c - x^*)], \quad x_c = \frac{q_c \rho}{N_c}, \quad (6.28)$$

where $k_{\text{au}} = 600 \text{ m}^3/(\text{kg s})$, $x^* = 2.6 \times 10^{-10} \text{ kg}$ is the separation mass, and the sigmoid onset replaces the hard threshold.

Self-collection and breakup.

$$\left. \frac{dN_r}{dt} \right|_{\text{sc}} = -k_{\text{sc}} N_r q_r \rho, \quad \left. \frac{dN_r}{dt} \right|_{\text{br}} = +k_{\text{br}} \sigma[\kappa(D_r - D_{\text{eq}})] \left| \left. \frac{dN_r}{dt} \right|_{\text{sc}} \right|. \quad (6.29)$$

6.4.3 Morrison Double-Moment (Ice + Liquid)

The Morrison two-moment scheme (Morrison et al., 2005) extends the warm-rain spectrum to mixed-phase clouds, adding prognostic cloud ice and snow and the ice-microphysical pathways that convert between them:

Ice nucleation (Cooper, 1986). The number of activated ice crystals rises exponentially as temperature falls below freezing, following the Cooper diagnostic:

$$N_{i,\text{target}} = \frac{N_{i,0}}{\rho} \exp[a_C(T_{\text{freeze}} - T)]. \quad (6.30)$$

Depositional growth.

$$\left. \frac{dq_i}{dt} \right|_{\text{dep}} = k_{\text{dep}} \text{softplus}(S_i) \sqrt{N_i}, \quad S_i = \frac{q_v}{q_{s,i}} - 1. \quad (6.31)$$

Bergeron process (mixed-phase). Cloud water converts to ice in the temperature window $T_{\text{center}} - T_{\text{width}} < T < T_{\text{freeze}}$:

$$\left. \frac{dq_c}{dt} \right|_{\text{Berg}} = -k_{\text{Berg}} q_c f_{\text{ice}}(T). \quad (6.32)$$

Riming, aggregation, melting. Riming: $q_c \rightarrow q_i$ and $q_c \rightarrow q_s$ via collection. Aggregation: $q_i \rightarrow q_s$ when crystals grow large enough. Melting: sigmoid-activated near T_{freeze} .

Fall speeds (power law).

$$V_t = a \left(\frac{\rho q}{\rho_{\text{sfc}}} \right)^b, \quad \text{rain: } a = 130, b = 0.5; \quad \text{ice: } a = 50, b = 0.25; \quad \text{snow: } a = 30, b = 0.3. \quad (6.33)$$

6.4.4 Thompson Hybrid-Moment Microphysics

The Thompson scheme (Thompson et al., 2008) adds a graupel category, predicting number for cloud ice and rain while diagnosing the remaining species. Rimed ice exceeding a critical threshold converts to graupel, the sigmoid gating the conversion smoothly above R_{thresh} :

$$\left. \frac{dq_g}{dt} \right|_{\text{rime}} = f_g \left. \frac{dq_i}{dt} \right|_{\text{rime}} \cdot \sigma \left[\kappa \left(\left. \frac{dq_i}{dt} \right|_{\text{rime}} - R_{\text{thresh}} \right) \right]. \quad (6.34)$$

6.4.5 Summary of Microphysics Schemes

Scheme	Moments	Species	Key features
Kessler	1	q_c, q_r	Warm rain only
Seifert–Beheng	2	q_c, q_r, N_c, N_r	Realistic droplet spectra
Morrison	2	$q_c, q_r, q_i, q_s, N_c, N_r, N_i$	Full mixed-phase
Thompson	Hybrid	$+q_g$	Graupel, riming

6.5 Cloud Fraction

6.5.1 Sundqvist (1988)

The Sundqvist closure (Sundqvist et al., 1989) diagnoses cloud fraction from grid-mean relative humidity, assuming subgrid moisture variability fills the cell with cloud as RH rises from a critical value RH_{crit} to saturation:

$$c_f = \text{clamp} \left(\frac{\text{RH} - \text{RH}_{\text{crit}}}{1 - \text{RH}_{\text{crit}}}, 0, 1 \right), \quad \text{RH}_{\text{crit}} = 0.8. \quad (6.35)$$

The clamp ensures $c_f \in [0, 1]$, with clear skies below threshold and full overcast at saturation.

6.5.2 Xu–Randall

Xu and Randall (1996) combine relative humidity and condensate amount, so that cloud fraction responds both to ambient RH and to the presence of condensate q_c :

$$c_f = \text{RH}^p \left[1 - \exp \left(- \frac{\alpha q_c}{(1 - \text{RH}) q_s} \right) \right], \quad p = 2, \alpha = 100. \quad (6.36)$$

Scheme dispatch. `cloud_fraction.compute_cloud_properties` dispatches on `config.scheme` and accepts only "sundqvist" or "xu_randall"; any other value raises `ValueError`. Disabling cloud diagnostics is handled *upstream* by setting `cloud_scheme="none"` on the caller's config so that `compute_cloud_properties` is not invoked at all — passing "none" to this function directly is itself an error. Earlier versions silently fell back to Sundqvist for any unrecognised string, masking config typos and dead branches; this is now uniform with the rest of the `integration.py` factory contract.

6.6 Boundary-Layer Turbulence

6.6.1 Louis (1979) Local Closure

The Louis scheme (Louis, 1979) is a first-order local closure: eddy diffusivities are set point-by-point from the local shear, mixing length, and stability, with no nonlocal transport. It is robust and inexpensive, which is why it remains a standard fallback above the boundary layer.

Mixing Length

The mixing length l interpolates between the surface-layer scaling $\kappa_{\text{vK}} z$ near the ground and an asymptotic ceiling l_{max} aloft, where κ_{vK} is the von Kármán constant:

$$l = \frac{\kappa_{\text{vK}} z}{1 + \kappa_{\text{vK}} z / l_{\text{max}}}, \quad \kappa_{\text{vK}} = 0.4, \quad l_{\text{max}} = 300 \text{ m}. \quad (6.37)$$

Richardson Number

The gradient Richardson number Ri measures the competition between stabilising buoyancy (N^2 , the squared Brunt–Väisälä frequency in virtual potential temperature θ_v) and destabilising shear (S^2); turbulence is suppressed where Ri is large and positive:

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2. \quad (6.38)$$

Stability Functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{Ri} / \sqrt{1 + d \text{Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (6.39)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

6.6.2 Holtslag–Boville Nonlocal K-Profile

Convective boundary layers transport heat and moisture even against the local gradient, which a purely local closure cannot capture. The nonlocal K-profile scheme (Holtslag and Boville, 1993) instead prescribes the diffusivity as a fixed-shape profile spanning the diagnosed PBL depth, optionally with a countergradient correction.

PBL Height Diagnosis

The PBL top is found from the bulk Richardson number, which compares the buoyancy deficit relative to the surface against the bulk wind shear:

$$\text{Ri}_{\text{bulk}}(z) = \frac{g}{\theta_{v,\text{sfc}}} \frac{(\theta_v - \theta_{v,\text{sfc}}) z}{|\mathbf{V}|^2}. \quad (6.40)$$

PBL height h is diagnosed where Ri_{bulk} crosses $\text{Ri}_{\text{crit}} = 0.3$ (LMD94 default), using sigmoid-weighted averaging for a smooth, differentiable diagnosis.

K-Profile Inside PBL

Within the PBL the diffusivity follows the cubic O'Brien-type profile: it scales with the friction velocity u_* and vanishes at both the surface and $z = h$, peaking in the mixed layer:

$$K_m = \kappa_{vK} u_* z (1 - z/h)^2. \quad (6.41)$$

Above the PBL, Louis (1979) Ri-dependent mixing applies.

6.6.3 TKE-Based Closure

Prognostic turbulent kinetic energy:

$$\frac{\partial \text{TKE}}{\partial t} = P - \varepsilon + D + B, \quad (6.42)$$

where P is shear production, $\varepsilon = c_\varepsilon \text{TKE}^{3/2}/l$ is dissipation, D is diffusion, and B is buoyancy production/destruction. Closure: $K_m = c_m \sqrt{\text{TKE}} \cdot l$, $K_h = K_m/\text{Pr}_t$.

6.6.4 Implicit Vertical Diffusion

The backward-Euler tridiagonal system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm:

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (6.43)$$

6.7 Surface Exchange Formulations

6.7.1 Neutral Bulk Aerodynamic Fluxes

Surface fluxes are parameterised by bulk aerodynamic formulae, in which each flux is the product of air density ρ , a transfer coefficient, the wind speed $|\mathbf{V}|$, and the air-surface difference of the transported quantity. Momentum stress (τ_x, τ_y) , sensible heat SH, and latent heat LH follow the same structure:

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (6.44)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad \text{LH} = \rho L_v C_E |\mathbf{V}| (q_s - q_a), \quad (6.45)$$

with $C_D = C_H = C_E = 1.5 \times 10^{-3}$. The neutral limit takes these coefficients as constants; the next section makes them stability-dependent.

6.7.2 Monin-Obukhov Similarity Theory (MOST)

Monin-Obukhov similarity theory (Monin and Obukhov, 1954) organises surface-layer turbulence around a single stability parameter $\zeta = z/L$, making the transfer coefficients functions of stability rather than constants.

Obukhov length. The Obukhov length L is the height at which buoyant and shear production of turbulence are comparable; it is built from the friction velocity u_* and the turbulent scales θ_* , q_* of temperature and humidity:

$$L = -\frac{u_*^3 T_v}{\kappa_{vK} g (\theta_* + 0.61 T_a q_*)}. \quad (6.46)$$

Businger–Dyer stability functions. The integrated stability corrections ψ_m, ψ_h modify the neutral logarithmic profile: convex unstable forms ($\zeta < 0$) enhance mixing while the linear stable branch ($\zeta \geq 0$) of [Businger et al. \(1971\)](#) suppresses it. For strongly stable conditions the [Beljaars \(1995\)](#) formulation prevents the fluxes from collapsing to zero:

$$\psi_m(\zeta) = \begin{cases} 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan x + \frac{\pi}{2}, & \zeta < 0, \quad x = (1 - 16\zeta)^{1/4}, \\ -5\zeta, & \zeta \geq 0, \end{cases} \quad (6.47)$$

$$\psi_h(\zeta) = \begin{cases} 2 \ln\left(\frac{1+y}{2}\right), & \zeta < 0, \quad y = (1 - 16\zeta)^{1/2}, \\ -5\zeta, & \zeta \geq 0. \end{cases} \quad (6.48)$$

Transfer coefficients. The coefficients then follow from the stability-corrected logarithmic profiles, with z_0 and z_{0t} the momentum and thermal roughness lengths:

$$C_D = \left[\frac{\kappa_{vK}}{\ln(z/z_0) - \psi_m(\zeta)} \right]^2, \quad C_H = \frac{\kappa_{vK}^2}{[\ln(z/z_0) - \psi_m(\zeta)][\ln(z/z_{0t}) - \psi_h(\zeta)]}. \quad (6.49)$$

Because ζ itself depends on the fluxes, this is a fixed-point problem. The iterative solver (5 iterations via `jax.lax.fori_loop`) is fully differentiable.

6.7.3 COARE 3.0 Roughness

Over ocean the momentum roughness is sea-state dependent, combining a wind-driven Charnock term with a viscous smooth-flow limit, following the COARE 3.0 algorithm ([Fairall et al., 2003](#)):

$$z_0 = \alpha_c \frac{u_*^2}{g} + 0.11 \frac{\nu}{u_*}, \quad z_{0t} = \min(1.15 \times 10^{-4}, 5.5 \times 10^{-5} \text{Re}_*^{-0.6}), \quad (6.50)$$

where $\alpha_c = 0.011$ is the Charnock parameter.

6.7.4 Large & Yeager (2004) Roughness

An alternative parameterises the neutral 10-m drag coefficient C_{DN} directly as a function of the neutral wind speed U_{10N} , following the CORE bulk formulae of [Large and Yeager \(2004\)](#):

$$C_{DN} \times 10^3 = \frac{2.7}{U_{10N}} + 0.142 + 0.0764 U_{10N}. \quad (6.51)$$

The three terms capture the low-wind viscous regime, a roughly constant mid-wind plateau, and the linear growth of drag in high winds.

6.8 Gravity Wave Drag

6.8.1 Orographic GWD (Lindzen 1981)

Stably-stratified flow over topography launches vertically-propagating gravity waves that carry momentum upward and deposit it where they break, decelerating the resolved flow. The launch stress follows the orographic drag closure of [McFarlane \(1987\)](#) and the saturation/wave-breaking criterion of [Lindzen \(1981\)](#).

Surface stress. The launched stress τ_0 scales with density, the Brunt–Väisälä frequency N , the horizontal wavenumber k , the squared obstacle height h_{topo}^2 , and the low-level wind U :

$$\tau_0 = \rho N k h_{\text{topo}}^2 U, \quad N = \sqrt{\frac{g}{\theta} \frac{d\theta}{dz}}. \quad (6.52)$$

Saturation criterion. As the wave amplitude grows with height it eventually overturns the local stratification. Waves break when the Froude number $\text{Fr} = U/(N h_{\text{topo}})$ drops below a critical value. Stress is capped at the saturation value $\tau_{\text{sat}} = \rho |U|^3 k/N$, so that the excess momentum flux is deposited locally.

Momentum deposition. The drag on the resolved flow is the vertical divergence of the wave stress: where τ decreases with height, momentum is delivered to the mean flow:

$$\left. \frac{du}{dt} \right|_{\text{GWD}} = -\frac{1}{\rho} \frac{d\tau}{dz}. \quad (6.53)$$

Frictional heating. Energy conservation requires that the kinetic energy lost to wave drag reappear as heat, recovered from the dot product of wind and momentum tendency:

$$\left. \frac{dT}{dt} \right|_{\text{GWD}} = -\frac{1}{c_p} \left(u \frac{du}{dt} + v \frac{dv}{dt} \right). \quad (6.54)$$

6.9 Held–Suarez Forcing

The idealised forcing benchmark of [Held and Suarez \(1994\)](#) replaces full physics with analytic relaxation, providing a reproducible test of a dry dynamical core’s general circulation.

Equilibrium temperature. The radiative-equilibrium target T_{eq} imposes an equator-to-pole contrast (ΔT_y) and a static-stability gradient ($\Delta \theta_z$), floored at 200 K to mimic the stratosphere:

$$T_{\text{eq}} = \max \left(200, \left[315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi \right] (p/p_0)^\kappa \right), \quad (6.55)$$

with $\Delta T_y = 60$ K, $\Delta \theta_z = 10$ K.

Newtonian relaxation. Temperature is restored toward T_{eq} at a rate k_T that is fast in the tropical boundary layer (mimicking strong surface coupling) and slow in the free atmosphere, the transition keyed to σ_b :

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max \left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b} \right) \cos^4 \phi, \quad (6.56)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction. Low-level momentum is damped to represent boundary-layer drag, again confined below σ_b :

$$\frac{dv}{dt} = -k_v(\sigma) v, \quad k_v = k_f \max \left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b} \right), \quad k_f = 1/(1 \text{ day}). \quad (6.57)$$

6.10 Additional Turbulence Schemes

6.10.1 Smagorinsky

Constant eddy diffusivity: $K_m = K_h = \text{const.}$

6.10.2 YSU (Yonsei University)

Nonlocal K-profile boundary-layer scheme with explicit entrainment at the PBL top, following [Holtslag and Boville \(1993\)](#) with enhanced counter-gradient terms.

6.10.3 CLUBB-lite

A reduced higher-order closure in the spirit of CLUBB ([Golaz et al., 2002](#)), carrying five prognostic second moments: $\overline{w'^2}$, $\overline{\theta_l'^2}$, $\overline{r_t'^2}$, $\overline{w'\theta_l'}$, $\overline{w'r_t'}$. Cloud fraction and condensate are diagnosed by integrating an assumed bivariate Gaussian PDF of θ_l and r_t , unifying boundary-layer turbulence and shallow-cloud closure in a single framework.

6.10.4 EDMF Turbulence

Eddy-diffusivity mass-flux unified boundary-layer framework combining diffusive transport (for weakly convective conditions) with explicit updraft mass flux (for strongly convective conditions) in a single scheme.

6.11 Additional Convection Schemes

6.11.1 Kuo

The [Kuo \(1974\)](#) column moisture-excess scheme activates when column-integrated moisture convergence exceeds a threshold. Excess moisture is partitioned between heating (fraction α_{heat}) and moistening ($1 - \alpha_{\text{heat}}$).

6.11.2 Deep Convective Adjustment (DCA)

Bottom-to-top column adjustment toward a moist-adiabatic reference profile, with smooth sigmoid activation for JAX differentiability.

6.12 Gravity Wave Drag Schemes

Six GWD parameterisations are available:

1. **Rayleigh friction:** Linear damping above a threshold height.
2. **Lindzen (1981):** Orographic GWD with saturation and critical-level absorption ([Lindzen, 1981](#)).
3. **McFarlane (1987):** Enhanced orographic scheme with improved wave saturation ([McFarlane, 1987](#)).
4. **Hines (1997):** Doppler-spread parameterisation for non-orographic GWD, propagating a spectrum of gravity waves upward with nonlinear interactions ([Hines, 1997](#)).

5. **Prognostic Spectral:** Multi-azimuthal (N_{az}), multi-wavenumber (N_k) spectrum with prognostic wave momentum flux and vertical propagation via `jax.lax.scan`, following the spectral non-orographic approach of [Alexander and Dunkerton \(1999\)](#).
6. **ML Emulator:** Equinox MLP surrogate mapping column state to GWD tendencies.

6.13 Neural and Learned Physics

Because every parameterisation above is differentiable, learned components can be substituted for, or blended with, the physical schemes and trained end-to-end through the dynamical core. This mirrors the hybrid neural–physical modelling strategy of [Kochkov et al. \(2024\)](#); related operator-learning work uses Fourier neural operators ([Li et al., 2021](#)) and their spherical variant ([Bonev et al., 2023](#)) for the global maps (see Chapter 12).

6.13.1 NeuralPhysics

An Equinox MLP column model that maps the column state (temperature, moisture, wind, surface fluxes) to physics tendencies. Hybrid mode blends learned corrections with conventional parameterisation output via a configurable blending coefficient.

6.13.2 LearnedColumn

Per-column physics replacement following [Rasp et al. \(2018\)](#), mapping the full column state vector to temperature and moisture tendencies via a neural network, trainable end-to-end with the dynamical core.

Chapter 7

Ocean Model

7.1 Boussinesq Hydrostatic Primitive Equations

The ocean dynamical core integrates the hydrostatic Boussinesq primitive equations in finite-volume form (Marshall et al., 1997), in which the full in-situ density is replaced by a constant reference density ρ_0 in the inertial terms while density variations are retained only in the buoyancy (pressure-gradient) term. Writing the horizontal momentum balance in vector-invariant form on the cubed sphere isolates the absolute vorticity and kinetic-energy gradient, which keeps the advection operator energy- and enstrophy-aware on the staggered grid:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (7.1)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (7.2)$$

Here u, v are the horizontal velocity components, ζ the relative vorticity, f the Coriolis parameter, $K = \frac{1}{2}(u^2 + v^2)$ the kinetic energy per unit mass, p' the baroclinic pressure perturbation, and A_h, A_v the lateral and vertical eddy viscosities; the final term $-w \partial u / \partial z$ is vertical advection of momentum. Each line therefore expresses the local acceleration as the sum of Coriolis, kinetic-energy gradient, pressure-gradient, lateral- and vertical-diffusion, and vertical-advection contributions. The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies, so that the fast surface-gravity-wave Coriolis rotation is treated at the short barotropic step.

7.1.1 Tracer Equations (Skew-Symmetric)

Each active and passive tracer ϕ (potential temperature, salinity, biogeochemical fields) obeys an advection–diffusion balance. The horizontal advection is written in skew-symmetric form, averaging the advective $-\mathbf{v} \cdot \nabla \phi$ and flux $-\nabla \cdot (\phi \mathbf{v})$ operators; because the velocity field is discretely non-divergent, this combination conserves both the tracer mean and its variance, suppressing the spurious extrema a one-sided form would generate:

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (7.3)$$

The remaining terms are vertical advection and the lateral (K_h) and vertical (K_v) diffusive fluxes.

7.1.2 Free-Surface Equation

Vertically integrating the continuity equation gives the prognostic free-surface height η , whose tendency is the column-summed horizontal volume-flux divergence:

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (7.4)$$

This sets the fast external-gravity-wave mode that the barotropic solver below integrates with a short substep.

7.2 Barotropic Solver

To avoid resolving the fast surface-gravity wave at the slow baroclinic timestep, the depth-integrated (barotropic) mode is advanced separately with a split-explicit forward–backward integrator ([Shchepetkin and McWilliams, 2005](#)), sub-cycled many short substeps Δt_s per baroclinic step. The barotropic state is the surface height η and the depth-mean transport $\bar{\mathbf{V}}$, forced by the slow tendency $\mathbf{F}_{\text{slow}}$ that carries the vertically integrated baroclinic pressure gradient and advection:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (7.5)$$

The forward–backward ordering (update η first, then use the new η in the momentum step) damps the computational mode and extends the stable substep relative to a centred scheme.

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (7.6)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$. Treating the Coriolis rotation implicitly over the substep prevents the inertial oscillation from being amplified by the many barotropic sub-cycles and keeps the substep stability set by the gravity-wave CFL alone.

7.3 Equation of State (Wright 1997)

The ocean prognostic temperature variable T is **potential temperature** [degC]. The Wright (1997) EOS ([Wright, 1997](#)) computes in-situ density $\rho(\theta, S, p)$ directly from potential temperature θ , salinity S [PSU], and pressure p [Pa]. This rational-function fit captures the leading nonlinearity of seawater density (thermobaricity and cabbeling), which drives the buoyancy term in the momentum equations; a higher-order TEOS-10 polynomial alternative ([Roquet et al., 2015](#); [McDougall and Barker, 2011](#)) is available where conservative temperature is the prognostic variable:

$$\rho = \frac{p + p_0(\theta, S)}{\lambda(\theta, S) + \alpha_0(\theta, S)(p + p_0)}, \quad (7.7)$$

where

$$\alpha_0 = a_0 + a_1 \theta + a_2 S, \quad (7.8)$$

$$p_0 = (b_0 + b_4 S) + \theta(b_1 + \theta(b_2 + b_3 \theta) + b_5 S), \quad (7.9)$$

$$\lambda = (c_0 + c_4 S) + \theta(c_1 + \theta(c_2 + c_3 \theta) + c_5 S). \quad (7.10)$$

Here α_0 , p_0 and λ are low-order polynomials in (θ, S) with coefficients a_i, b_i, c_i . Coefficients follow the MOM6 implementation. Only the density perturbation $\rho' = \rho(\theta, S, p) - \rho_0$ enters the pressure-gradient computation, consistent with the Boussinesq approximation in which ρ_0 carries the inertia and ρ' supplies the buoyancy.

7.4 Hydrostatic Pressure and Vertical Velocity

Under the hydrostatic approximation the vertical momentum equation reduces to the balance between the vertical pressure gradient and buoyancy, so the baroclinic pressure at level k follows from downward integration of the density perturbation, with the surface load $\rho_0 g \eta$ as the upper boundary value and a half-cell term that centres the pressure within the k -th layer:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (7.11)$$

The vertical velocity is not prognostic; it is diagnosed from continuity by integrating the horizontal-flux divergence upward from the rigid floor ($w = 0$ at the bottom), so that the discrete velocity field remains non-divergent layer by layer:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (7.12)$$

7.5 Vertical Mixing: K-Profile Parameterisation (KPP)

Following [Large et al. \(1994\)](#):

7.5.1 Boundary Layer Depth

The boundary layer depth h is diagnosed where the bulk Richardson number crosses $\text{Ri}_{\text{crit}} = 0.3$ (LMD94 default):

$$\text{Ri}_b(z) = \frac{g \Delta \rho z}{\rho_0 (|\Delta \mathbf{V}|^2 + V_t^2)}, \quad (7.13)$$

using sigmoid-weighted averaging for a differentiable diagnosis.

7.5.2 K-Profile Shape

Within the diagnosed boundary layer the eddy diffusivity is prescribed as a smooth cubic shape function $G(\sigma)$ scaled by the layer depth h and the turbulent velocity scale w_s , where $\sigma = z/h$ is the fractional depth. The shape vanishes at the surface and at the layer base and peaks in the interior, matching the observed near-surface turbulent diffusivity profile:

$$K_m(\sigma) = h w_s(\sigma) G(\sigma), \quad G(\sigma) = \sigma(1 - \sigma)^2, \quad \sigma = z/h, \quad (7.14)$$

where w_s is the turbulent velocity scale depending on surface fluxes and stability.

7.5.3 Non-Local Transport

For unstable (convective) boundary layers ($B_f > 0$), a non-local counter-gradient flux is added to temperature and salinity tendencies. The implementation uses a simplified form inspired by LMD94 Eq. 19:

$$\left. \frac{\partial T}{\partial t} \right|_{\text{NL}} = \gamma_T \frac{w_s(0) G(\sigma)}{w_s(\sigma) h} \left. \frac{\partial T}{\partial z} \right|_{\text{sfc}}, \quad (7.15)$$

where $\gamma_T = \gamma_S = 6.33$ are non-local transport coefficients and $w_s(0) (\partial T / \partial z)|_{\text{sfc}}$ is a surface heat flux proxy [K m/s]. This differs from the full LMD94 non-local term, which uses the explicit surface buoyancy flux $w'\theta'_0$; the present form approximates it via the near-surface gradient and surface turbulent velocity scale. This is active only within the diagnosed boundary layer ($\sigma < 1$) and only under unstable forcing.

7.6 Lateral Mixing

Lateral mixing supplies the subgrid closure for unresolved horizontal stirring and, in eddy-permitting configurations, the scale-selective dissipation that controls the grid-scale noise the advection scheme cannot.

7.6.1 Harmonic (Laplacian) Diffusion

The simplest closure is a constant-coefficient Laplacian diffusion, which damps all scales monotonically:

$$D_h = A_h \nabla^2 \phi. \quad (7.16)$$

7.6.2 Biharmonic (∇^4) Diffusion

A biharmonic operator is more scale-selective: it leaves the resolved large scales nearly untouched while strongly damping the $2\Delta x$ computational mode, which is preferred for eddy solutions. The coefficient A_4 may be set by a Smagorinsky-like strain-rate closure following [Griffies and Hallberg \(2000\)](#):

$$D_h = -A_4 \nabla^4 \phi. \quad (7.17)$$

7.6.3 Gent–McWilliams / Redi Parameterisation

The GM parameterisation represents the effect of mesoscale eddies on tracer transport via an additional advective flux (bolus velocity) ([Gent and McWilliams, 1990](#)):

$$\mathbf{u}^* = -\kappa_{\text{GM}} \nabla_h s, \quad (7.18)$$

where $s = -\nabla_h \rho / (\partial \rho / \partial z)$ is the isopycnal slope and κ_{GM} the thickness diffusivity. This adiabatic eddy transport flattens isopycnals and extracts available potential energy, mimicking baroclinic-eddy restratification. The accompanying Redi diffusion tensor ([Redi, 1982](#)) rotates the lateral diffusion to act along isopycnal surfaces rather than along geopotentials, which suppresses the spurious diapycnal mixing a horizontal Laplacian would otherwise introduce on sloping density surfaces.

7.7 Tracer Advection Schemes

The legoESM ocean carries a family of flux-form tracer advection schemes of increasing accuracy in `src/legoesm/ocean/advection.py` and `advection_som.py`. Following Hill et al. (2012), implicit numerical mixing is the dominant source of spurious diapycnal transport in z -coordinate models, so the volume-weighted variance $\text{Var}(T) = \int (T - \langle T \rangle)^2 dV$ is used as the definitive evaluation metric (it is an exact inviscid invariant in closed-domain zero- K_h Eady setups).

Scheme inventory.

Scheme	Order / halo	Notes
<code>upwind</code>	1, $h=1$	Donor-cell; most diffusive but unconditionally stable.
<code>tvb</code>	2, $h=2$	Van Leer / Sweby slope-limited flux (van Leer, 1979); total-variation diminishing.
<code>dst3</code>	3, $h=2$	Direct-space-time third-order, used in vertical advection.
<code>ppm</code>	4, $h=2$	Parabolic Piecewise Method (Colella and Woodward, 1984).
<code>ppm_fct</code>	4, $h=2$	PPM with Zalesak FCT limiter (sign-split limiter pending, #9).
<code>weno5/weno7</code>	5/7, $h=3/4$	WENO smoothness-weighted reconstructions (Jiang and Shu, 1996).
<code>som</code>	Prather 9-moment	Second-Order Moments (Prather, 1986); advects all 9 sub-cell moments per tracer.

SOM (Prather 1986). The second-order-moment scheme (Prather, 1986) attains very low numerical diffusion by transporting the sub-cell tracer distribution explicitly rather than reconstructing it each step. The SOM solver in `advection_som.py` carries the cell mean and nine sub-cell moments $\{S_x, S_y, S_z, S_{xx}, S_{yy}, S_{zz}, S_{xy}, S_{xz}, S_{yz}\}$, performs operator-split donor/receiver sweeps (`_som_x_sweep`, `_som_y_sweep`, `_som_z_sweep`), and applies moment limiters (`_limit_moments`) to enforce monotonicity. The scheme is unconditionally monotone and conserves all advected moments to machine precision; it is the lowest-numerical-diffusion option in the inventory.

Eady channel benchmark. On the Eady uniform channel (100×50 , 20 levels, no sponge, $B_h = 2.3 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$, $K_h = A_h = 0$):

Scheme	200 d strong ($U=0.8$)	600 d weak ($U=0.2$)	Cost vs. <code>upwind</code>
<code>upwind</code>	86% $\text{Var}(T)$ lost	12% lost	1.0×
<code>tvb</code>	70% lost	6% lost	1.02×
<code>som</code>	blow-up day 107	—	2.8×
<code>som + C_{Smag}=0.2</code>	46% lost	2.9% lost	2.2×
<code>ppm_fct</code>	blow-up day 30	—	~1.3× (est.)

The strong-forcing regime saturates at $p_{95}|\text{Ro}| \approx 1.5\text{--}3$ (submesoscale, where diapycnal mixing is physically mandatory); the weak-forcing regime stays at $p_{95}|\text{Ro}| \approx 0.05\text{--}0.11$ (balanced ocean interior), so even `upwind` preserves 88.5% of $\text{Var}(T)$ over 600 days. The *absolute* amount of numerical mixing is dominated by the forcing regime, not the scheme.

Recommendations. Production AMIP / coupled runs default to `tvb` (safe, $\leq 2\%$ overhead); science runs targeting fronts and water-mass tracking use `som+Smagorinsky` ($C_{\text{Smag}}=0.2$) which combines nine-moment conservation with just enough momentum dissipation to suppress grid-scale noise. The `ppm_fct` branch is gated on the sign-split Zalesak/Kuzmin limiter (open issue #9). Convergence methodology and 2D/3D test runners are documented in `docs/ocean_experiments/advection_convergence`.

7.8 Slab and Two-Layer Ocean

For idealised experiments that do not require the full dynamical ocean, a slab mixed layer evolves the surface temperature T_s from the column energy balance: net shortwave and longwave radiation, sensible (SH) and latent (LH) turbulent fluxes, a prescribed ocean-heat-transport convergence Q_{flux} that closes the climatological energy budget, and a restoring exchange $K_{\text{mix}}(T_{\text{deep}} - T_s)$ with the deep reservoir:

$$C_{\text{mix}} \frac{dT_s}{dt} = \text{SW}_{\text{net}} + \text{LW}_{\text{net}} - \text{SH} - \text{LH} + Q_{\text{flux}} + K_{\text{mix}}(T_{\text{deep}} - T_s), \quad (7.19)$$

where $C_{\text{mix}} = \rho_{\text{ocn}} c_{\text{ocn}} h_{\text{mix}}$ is the mixed-layer heat capacity set by the mixed-layer depth h_{mix} , which fixes the thermal inertia and hence the seasonal SST response.

7.9 Realistic-Geometry Spinup on ETOPO Bathymetry

The lat-lon C-grid ocean supports global Wolfe–Cessi-style spinups on real ETOPO/GEBCO bathymetry through a topology-aware loader and a set of stability-conditioning operators specific to coastal partial cells and high latitudes.

Bathymetry pre-processing. `src/legoesm/ocean/bathymetry.py` exposes `BathymetryConfig` and the per-grid loaders (`load_bathymetry_latlon_cgrid`, `load_bathymetry_cubed_sphere`, `load_bathymetry_gaussian`). The pipeline is:

1. Auto-detect the depth/elevation variable across ETOPO, GEBCO, NOAA ERDDAP, and generic CF conventions.
2. Regrid to the model grid (conservative or bilinear) and clip to $H \in [H_{\text{min}}, H_{\text{max}}]$ with $H_{\text{min}} = 50$ m by default (preserves mid-ocean ridges).
3. Optional Laplacian smoothing on the cubed-sphere or Voronoi mesh (`_laplacian_smooth_2d`, `_laplacian_smooth_voronoi`).
4. Steepness-cap via the Mellor–Ezer–Oey iteration `apply_meo_r_factor_cap` which enforces $r_{\text{max}} = |\Delta H|/(2\bar{H}) \leq r_{\text{cap}}$ (typical $r_{\text{cap}} = 0.2$); the per-call `initial_r_max` / `final_r_max` are logged for the diagnostic scripts.
5. Optional polar-cap mask (`north_cap_lat`, `south_cap_lat`) and `enforce_straits` to thread Drake / Indonesian throughflow at low resolution.
6. Isolated-basin filling (`_fill_isolated_basins`) so disconnected pockets do not trap tracers.

AL81 partial-cell Coriolis. The vector-invariant momentum branch uses an Arakawa–Lamb 1981 (“EEN”) 12-corner-triad PV-flux closure (`pv_flux_al81_partial_cell` in `latlon_cgrid_operators.py`). Each f -point carries four 1/12-weighted L-shaped triads of vertex potential vorticity $q = \zeta/h_{\text{vtx}}$; the partial-cell extension after [Le Sommer et al. \(2009\)](#) weights each triad by the local wet-cell thickness so that discrete potential enstrophy is preserved across bottom steps. Coastal triads use a Neumann fill of q (three-pass smoothing band-aid pending the formal mask-projection closure documented in `docs/ocean_experiments/al81_corner_triad_audit.md`).

Biharmonic viscosity scaling. Because lat-lon zonal spacing scales as $\cos \phi$, a constant biharmonic coefficient B_h violates CFL near the poles while under-diffusing at the equator. `biharmonic_scaling_factor` multiplies B_h by $\cos^4 \phi$ at u - and v -faces following the MOM6 convention of Griffies and Hallberg (2000), so the effective diffusivity tracks dx_{local}^4 uniformly. This is the dominant lever for suppressing the alternating-sign $2\Delta y$ zonal-jet computational mode that develops at the equator on partial cells.

Validated configuration. The 20-yr `global_overturning_realistic_geometry` run (branch `realistic-geometry-full`, PR #224) uses $A_h = 10^4 \text{ m}^2 \text{ s}^{-1}$, $B_h = 5 \times 10^9 \text{ m}^4 \text{ s}^{-1}$, $\kappa_{\text{GM}} = \kappa_{\text{Redi}} = 800 \text{ m}^2 \text{ s}^{-1}$, SMC03 pressure-gradient force, AL81 vorticity advection, implicit Crank–Nicolson barotropic solver, and bottom-drag $r = 2.5 \times 10^{-3}$. Stratification reproduces the canonical thermohaline structure (warm pool aloft, cold abyss, sloping isotherms, warm-equator/cold-pole SST imprint). Surface speeds remain inflated $\sim 3\text{--}5\times$ above realistic values from residual partial-cell q -stencil noise; the follow-up plan is the full Sadourny–Salmon energy–enstrophy form with all bottom-cell ENE corrections. See `docs/ocean_experiments/realistic_geometry_phase4_results.md` for the year-by-year diagnostics.

Chapter 8

Ocean Biogeochemistry

8.1 Carbonate Chemistry and Air–Sea CO₂ Exchange

8.1.1 Gas Transfer Velocity

Air–sea CO₂ exchange is parameterised as a bulk flux proportional to a gas transfer (piston) velocity k_w , which the quadratic wind-speed relation of Wanninkhof (2014) ties to the 10 m wind U_{10} and the CO₂ Schmidt number Sc , normalised to its reference value of 660:

$$k_w = 0.251 U_{10}^2 \left(\frac{Sc}{660} \right)^{-1/2}, \quad (8.1)$$

where the Schmidt number is a polynomial function of temperature:

$$Sc = 2116.8 - 136.25 T + 4.7353 T^2 - 0.092307 T^3 + 7.555 \times 10^{-4} T^4. \quad (8.2)$$

8.1.2 Net Air–Sea Flux

The net flux is the piston velocity times the air–sea difference in CO₂ partial pressure, converted to a concentration gradient by the temperature- and salinity-dependent Henry’s-law solubility K_0 ; the sign convention makes $F_{CO_2} > 0$ an outgassing from a supersaturated ocean:

$$F_{CO_2} = k_w K_0 (pCO_{2,ocean} - pCO_{2,atm}), \quad (8.3)$$

where K_0 is Henry’s law solubility. The ocean surface pCO_2 is computed from DIC, ALK, T , S via the carbonate equilibria, solved for $[H^+]$ with the Newton–Raphson iteration of Follows et al. (2006), which exploits the near-linear dependence of total alkalinity on H^+ to converge in a few steps within the JIT-compiled column loop.

8.2 NPZD Ecosystem Model

The ecosystem closure carries four biological tracers, NO₃ (nutrients), P (phytoplankton), Z (zooplankton), and D (detritus), exchanging nitrogen along a single trophic chain following Fasham et al. (1990). Each budget balances the source and sink terms of the cycle: phytoplankton gain by growth $\mu_P P$ and lose to grazing G and mortality; grazed nitrogen is partitioned between assimilated zooplankton biomass (efficiency γ_Z) and egested detritus; detritus remineralises at rate r_D

and sinks at speed w_s ; and nitrate is drawn down by uptake and replenished by remineralisation, closing the nitrogen budget:

$$\frac{\partial P}{\partial t} = \mu_P P - G - m_P P, \quad (8.4)$$

$$\frac{\partial Z}{\partial t} = \gamma_Z G - m_Z Z^2, \quad (8.5)$$

$$\frac{\partial D}{\partial t} = (1 - \gamma_Z)G + m_P P + m_Z Z^2 - r_D D + w_s \frac{\partial D}{\partial z}, \quad (8.6)$$

$$\frac{\partial \text{NO}_3}{\partial t} = -\mu_P P + r_D D. \quad (8.7)$$

Phytoplankton growth. The growth rate multiplies the maximum rate $\mu_{\max}$ by a temperature factor $f(T)$ and the more limiting of nutrient and light availability (Liebig's law of the minimum), with the nutrient limitation a Michaelis–Menten saturation in NO_3 and the light limitation a saturating function of photosynthetically available radiation (PAR):

$$\mu_P = \mu_{\max} f(T) \min(L_N, L_I), \quad L_N = \frac{\text{NO}_3}{k_N + \text{NO}_3}, \quad L_I = 1 - e^{-k_{\text{PAR}} \cdot \text{PAR}}. \quad (8.8)$$

Grazing (Holling III). Zooplankton grazing uses a Holling type-III (sigmoidal) functional response, whose P^2 numerator gives a low-prey refuge that stabilises the predator–prey oscillation:

$$G = g_{\max} \frac{P^2}{K_g + P^2} Z. \quad (8.9)$$

Redfield stoichiometry ($R_{C:N} = 6.625$) couples biology to DIC/ALK.

8.3 Ocean Bottom Drag

The bottom stress τ_b removes momentum from the lowest wet layer and is the main sink for the barotropic flow; two formulations are available.

Linear drag. A Rayleigh (linear) drag $\tau_b = -r \mathbf{u}_b$, with configurable coefficient r , is the simplest closure and keeps the momentum operator linear for idealised spin-down problems.

Quadratic drag. The quadratic law $\tau_b = -C_d |\mathbf{u}_b| \mathbf{u}_b$, with $C_d = 2.5 \times 10^{-3}$ (default), follows the turbulent log-layer scaling and is preferred for realistic geometry where bottom speeds vary widely.

8.4 Ocean Surface Forcing

The surface boundary condition supplies the momentum, heat, and freshwater fluxes that drive the ocean. Three modes are available, trading physical fidelity against simplicity:

- **Prescribed:** Fixed wind stress (τ_x, τ_y) and heat flux with latitudinal profiles.
- **Restoring:** SST/SSS restoring toward climatology (Haney-style) with cosine latitude temperature profile (25°C at equator, 0°C at poles).

- **Bulk formulas:** COARE-like air–sea fluxes with drag (C_D), Stanton (C_H), and Dalton (C_E) coefficients, reference height, and stability corrections, in the CORE forced-ocean configuration of [Large and Yeager \(2004\)](#).

The bulk mode lets the ocean feed back on its own fluxes (a warmer surface reduces sensible/latent loss), which a prescribed-flux setup cannot, and is the standard choice for hindcast spinups.

8.5 Shortwave Penetration

Solar radiation that penetrates the surface heats a finite layer rather than just the top cell, which controls the near-surface stratification and mixed-layer heat budget. Subsurface shortwave absorption follows the Jerlov water-type classification ([Jerlov, 1976](#)) with five types (I, IA, IB, II, III). Each type defines a two-band exponential decay profile ([Paulson and Simpson, 1977](#)), splitting the flux into a rapidly attenuated red band and a more penetrating blue-green band:

$$I(z) = I_0 [R e^{-z/\zeta_1} + (1 - R) e^{-z/\zeta_2}], \quad (8.10)$$

where the partition R and the two e-folding depths ζ_1, ζ_2 depend on the water type (clearer water gives larger ζ_2 and deeper heating).

8.6 Ocean Convection

Static instability ($N^2 < 0$) must be removed rapidly to represent deep convective overturning; two closures are provided.

8.6.1 Enhanced Diffusion

The simplest is convective adjustment by enhanced diffusivity: wherever the stratification is statically unstable ($N^2 < 0$), the vertical diffusivity is raised to $K_{\text{conv}} = K_{\text{max}}$, with smooth (sigmoid) transitions to keep the closure differentiable, so that the unstable column mixes back toward neutral within a few steps.

8.6.2 Plume Model

A more physical alternative is an entraining mass-flux plume that carries dense surface water downward, with entrainment rate $\alpha_{\text{plume}} = 0.1$ diluting the plume with ambient water and a background detrainment $\varepsilon = 10^{-3} \text{ m}^{-1}$ depositing it, resolving the convective transport rather than only diffusing the instability away.

8.7 Richardson-Number Vertical Mixing

Below the surface boundary layer, mixing is driven by shear instability, parameterised so that the diffusivity decreases monotonically with the gradient Richardson number $\text{Ri} = N^2/S^2$, the ratio of stabilising buoyancy stratification N^2 to destabilising vertical shear S^2 :

$$K_v = K_{v,0} (1 + 5 \text{Ri})^{-1}, \quad \text{Ri} = \frac{N^2}{S^2}, \quad (8.11)$$

so that low-Ri (strongly sheared) regions mix toward the maximum $K_{v,0}$ while well-stratified, weakly sheared interior layers approach the molecular background. This complements the KPP boundary-layer scheme of Section 7.5 with an interior shear-instability closure.

Chapter 9

Land Surface Model

9.1 Slab Land

9.1.1 Surface Energy Balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) \text{SW}^\downarrow + \varepsilon_l (\text{LW}^\downarrow - \sigma_{\text{SB}} T_s^4) - \text{SH} - \text{LH}. \quad (9.1)$$

This is the surface energy balance for a slab of heat capacity $C_s d_s$: absorbed shortwave $(1 - \alpha_l) \text{SW}^\downarrow$ (with albedo α_l), net longwave (downwelling $\text{LW}^\downarrow$ minus emission $\varepsilon_l \sigma_{\text{SB}} T_s^4$ at emissivity ε_l), and turbulent sensible SH and latent LH losses sum to the storage tendency. The finite slab heat capacity gives the surface a diurnal memory rather than forcing instantaneous radiative equilibrium.

9.1.2 Bucket Hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\text{max}}]. \quad (9.2)$$

Moisture availability $\beta = \beta_{\text{min}} + (1 - \beta_{\text{min}}) W/W_{\text{max}}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

9.2 Multi-Layer Land Model

9.2.1 Soil Thermal Diffusion

$$C_{\text{eff}}(z) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[k_{\text{eff}}(z) \frac{\partial T}{\partial z} \right], \quad (9.3)$$

with effective heat capacity:

$$C_{\text{eff}} = (1 - \theta_{\text{sat}}) C_{\text{soil}} + \theta C_{\text{water}} + (\theta_{\text{sat}} - \theta) C_{\text{air}}, \quad (9.4)$$

and thermal conductivity following the [Johansen \(1975\)](#) parameterisation via the Kersten number. Solved implicitly using the Thomas algorithm with surface heat flux and geothermal flux boundary conditions.

9.2.2 Richards Equation (Soil Hydraulics)

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right], \quad (9.5)$$

where θ is volumetric water content, ψ is matric potential, and $K(\psi)$ is hydraulic conductivity. This is the mixed form of the Richards equation: vertical water flux is driven by the sum of the matric-potential gradient $\partial\psi/\partial z$ and the unit gravitational gradient, so that under saturated, hydrostatic conditions ($\partial\psi/\partial z = -1$) the flux vanishes.

The closure relating ψ and K to θ requires a water-retention curve. Six are available; the default van Genuchten form (Van Genuchten, 1980) gives a smooth, differentiable $\psi(\theta)$ with a continuous derivative at saturation, which is important for stable implicit solves and for autodiff:

Curve	$\psi(\theta)$
Van Genuchten (1980)	$\psi = \alpha^{-1}[(S_e^{-1/m} - 1)^{1/n}]$
Campbell (1974)	$\psi = \psi_s (S_e)^{-b}$
Brooks–Corey (1964)	$\psi = \psi_d (S_e)^{-1/\lambda}$
Clapp–Hornberger (1978)	Power-law form

where $S_e = (\theta - \theta_r)/(\theta_{\text{sat}} - \theta_r)$ is effective saturation.

Root water uptake. Root distribution: $f_{\text{root}}(z) = \exp(-z/z_{\text{root}})$, with default $z_{\text{root}} = 0.5$ m. Plant-available water between wilting point (θ_{wp}) and field capacity (θ_{fc}).

9.2.3 Snow Budget

$$\frac{dh_{\text{swe}}}{dt} = P_{\text{snow}} - M, \quad M = \max\left(0, \frac{Q_{\text{net}}}{L_f \rho_w}\right), \quad (9.6)$$

with age-dependent albedo: $\alpha_{\text{snow}}(t) = \alpha_{\text{max}} e^{-t/\tau_{\text{age}}} + \alpha_{\text{min}}(1 - e^{-t/\tau_{\text{age}}})$.

9.3 Land Carbon Cycle

9.3.1 DALEC-990 Six-Pool Model

Six prognostic pools: labile C_{lab} , foliage C_f , root C_r , wood C_w , litter C_{lit} , and soil organic matter C_{SOM} .

Gross Primary Production (GPP).

$$\text{GPP} = \varepsilon \text{APAR} f_T f_{\text{CO}_2} \beta, \quad \text{APAR} = f_{\text{APAR}} \text{PAR}, \quad f_{\text{APAR}} = 1 - e^{-k_{\text{ext}} \text{LAI}}, \quad (9.7)$$

where ε is light-use efficiency, $f_T = \exp[-0.5((T - T_{\text{opt}})/T_{\text{width}})^2]$ is the Gaussian temperature response, and $f_{\text{CO}_2} = \text{CO}_2/(K_{\text{CO}_2} + \text{CO}_2)$ is Michaelis–Menten CO_2 response.

Soil respiration. Exponential Q_{10} temperature sensitivity with moisture limitation.

9.3.2 Farquhar (1980) Photosynthesis

The biochemical C3 photosynthesis model of [Farquhar et al. \(1980\)](#) represents net assimilation as the slower of two co-limiting processes minus respiration:

$$A = \min(W_c, W_j) - R_d, \quad (9.8)$$

where A is net assimilation, W_c the Rubisco-limited and W_j the light- (RuBP-regeneration-)limited gross rate, and R_d leaf dark respiration:

$$W_c = V_{c,\max} \frac{C_i - \Gamma^*}{C_i + K_c(1 + O/K_o)}, \quad (\text{Rubisco-limited}), \quad (9.9)$$

$$W_j = \frac{J(C_i - \Gamma^*)}{4(C_i + 2\Gamma^*)}, \quad (\text{light-limited}). \quad (9.10)$$

Here C_i is the intercellular CO_2 , Γ^* the CO_2 compensation point, K_c and K_o the Michaelis constants for carboxylation and oxygenation, O the oxygen concentration, and J the electron-transport rate. The maximum carboxylation capacity $V_{c,\max}$ and $J_{\max}$ follow peaked Arrhenius temperature responses ([Bernacchi et al., 2001](#)), capturing the decline of enzyme activity at high leaf temperature. The co-limitation minimum is implemented as a smooth (differentiable) function so the assimilation rate remains gradient-friendly through the transition between the two regimes.

9.3.3 Stomatal Conductance

Three models couple stomata to photosynthesis. The first two are empirical optimality relations in which conductance g_s scales with assimilation A and a humidity term, with g_0 the residual (cuticular) conductance and g_1 the slope; closing the loop with the diffusion constraint $A = g_s(C_s - C_i)/1.6$ ties the carbon and water exchange consistently.

Ball–Berry (1987). $g_s = g_0 + g_1 A \text{RH}/C_s$, with conductance proportional to relative humidity RH at the leaf surface ([Ball et al., 1987](#)).

Medlyn (2011). $g_s = g_0 + 1.6(1 + g_1/\sqrt{\text{VPD}}) A/C_s$, derived from stomatal optimisation theory and expressed through vapour-pressure deficit VPD rather than relative humidity ([Medlyn et al., 2011](#)).

Jarvis (1976). $g_s = g_{s,\max} f(\text{PAR}) f(T) f(\text{VPD}) f(\theta)$, a multiplicative environmental-stress reduction of a maximum conductance (CO_2 -independent; for when the carbon cycle is inactive), following the [Jarvis \(1976\)](#) formulation.

The coupled A – g_s – C_i system is solved by 5 fixed-point iterations, unrolled for JAX JIT compatibility.

9.4 Plant Functional Types

Land surface parameters are spatially heterogeneous and depend on vegetation type. legoESM implements a CLM5-style Plant Functional Type (PFT) system with 17 types spanning needle-leaf/broadleaf trees (evergreen and deciduous, temperate and tropical), shrubs, grasses (C3/C4), and crops.

9.4.1 Parameter Lookup

For each grid cell, the effective parameter $\bar{p}$ is a PFT-fraction-weighted average:

$$\bar{p} = \sum_{k=1}^{17} f_k p_k, \quad (9.11)$$

where f_k is the fractional coverage of PFT k and p_k is the parameter value for that type.

9.4.2 Trainable Parameters

Twelve land surface parameters are trainable per PFT: vegetation albedo (α_{veg}), emissivity (ε), roughness length (z_0), maximum soil water (W_{max}), soil heat capacity (C_{soil}), soil depth (d_{soil}), root depth (z_{root}), wilting point (θ_{wp}), field capacity (θ_{fc}), maximum carboxylation rate ($V_{c,\text{max},25}$), leaf carbon mass per area (LCMA), and stomatal slope (g_1).

Each parameter is bounded via sigmoid transforms:

$$p = p_{\text{min}} + (p_{\text{max}} - p_{\text{min}}) \sigma(x_{\text{raw}}), \quad (9.12)$$

where x_{raw} is the unconstrained trainable value.

9.4.3 Neural Parameter Provider

An alternative to PFT lookup, a 3-hidden-layer fully connected network (GELU activation, 48 hidden units) maps static land features—cyclic latitude/longitude, one-hot soil type, normalised elevation, precipitation, temperature, and forest age—to the 12 land parameters. Sigmoid output bounds match the PFT physical ranges.

9.5 DifferLand: Differentiable Land Model Training

The DifferLand framework enables end-to-end gradient-based optimisation of land surface and carbon cycle parameters. Three carbon cycle modes are available:

- **none:** Carbon cycle disabled; LAI prescribed.
- **differland:** Prognostic DALEC-990 with trainable allocation fractions, decomposition rates, and phenology parameters.
- **seasonal:** Prescribed seasonal LAI cycle; carbon pools diagnostic only.

9.5.1 Phenology

Leaf emergence and fall follow hemisphere-aware Gaussian windows:

$$f_{\text{on}}(d) = \exp\left[-\frac{(d - d_{\text{Bday}})^2}{2\tau_{\text{on}}^2}\right], \quad f_{\text{off}}(d) = \exp\left[-\frac{(d - d_{\text{Fday}})^2}{2\tau_{\text{off}}^2}\right], \quad (9.13)$$

where d is the day of year, d_{Bday} and d_{Fday} are budburst and leaf-fall dates, and τ controls the transition width.

9.5.2 Decomposition

Litter and SOM decomposition rates follow Q_{10} temperature sensitivity with moisture limitation:

$$k(T, \theta) = k_0 Q_{10}^{(T - T_{\text{ref}})/10} f_w(\theta), \quad (9.14)$$

where f_w is a piecewise-linear function of soil moisture.

9.5.3 NPP Allocation

Net primary production ($\text{NPP} = \text{GPP} - R_a$) is allocated sequentially: foliar $\rightarrow$ labile $\rightarrow$ root $\rightarrow$ wood, with allocation fractions as trainable parameters.

Chapter 10

Sea Ice

10.1 Thermodynamic Slab

The thermodynamics follow the zero-layer slab tradition of [Semtner \(1976\)](#), with the energy-conserving enthalpy refinements of [Bitz and Lipscomb \(1999\)](#).

10.1.1 Surface Temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i(T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\text{min}}). \quad (10.1)$$

The ice surface temperature T_i responds to the net surface flux Q_{sfc} balanced against conduction of heat from the freezing ocean interface at T_f through the ice of effective thickness h_{eff} with conductivity k_i . The floor h_{min} keeps the conductive flux finite for vanishingly thin ice.

10.1.2 Growth and Melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (10.2)$$

The ice thickness h_i grows when the conductive flux F_{cond} extracting heat upward exceeds the ocean heat flux F_{ocean} supplied from below; the imbalance is converted to mass through the latent heat of fusion L_f . The sign convention therefore ties freezing and melting directly to the column energy budget.

10.1.3 Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1 - c)/h_{\text{new}} & \text{growing,} \\ (dh/dt) c/h_{\text{eff}} & \text{melting.} \end{cases} \quad (10.3)$$

10.2 EVP Rheology

Following the elastic–viscous–plastic (EVP) rheology of [Hunke and Dukowicz \(1997\)](#), the ice momentum equation is

$$m \frac{\partial \mathbf{u}_i}{\partial t} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_w - mf \hat{\mathbf{k}} \times \mathbf{u}_i - mg \nabla \eta + \nabla \cdot \boldsymbol{\sigma}, \quad (10.4)$$

where m is the ice mass per unit area, $\mathbf{u}_i$ the ice velocity, $\boldsymbol{\tau}_a$ and $\boldsymbol{\tau}_w$ the air and ocean stresses, f the Coriolis parameter, η the sea-surface height, and $\boldsymbol{\sigma}$ the internal stress tensor from an elliptic yield curve with eccentricity $e = 2$. The plastic limit is the physical target; the elastic term is a numerical regularisation that lets the equation be advanced by explicit subcycling rather than an implicit solve. Ice strength:

$$P = P^* h \exp(-C(1-c)), \quad P^* = 27.5 \text{ kPa}, \quad C = 20. \quad (10.5)$$

Here P^* is the ice-strength parameter and C the concentration-decay constant, so the plastic strength rises with thickness h and falls as the pack opens ($c \rightarrow 0$). The elastic term provides subcycling stability ($N_{\text{evp}} = 120$ substeps); the modified (mEVP) formulation (Bouillon et al., 2013) recasts the relaxation so that the subcycled solution converges to the same viscous-plastic limit independently of the substep count.

10.3 Multi-Category Ice Thickness Distribution

The subgrid ice-thickness distribution theory of Thorndike et al. (1975) treats the ice pack as a probability distribution over thickness rather than a single mean floe, which is essential because growth, melt, and the conductive flux are all strongly nonlinear in thickness. For $N_{\text{cat}} > 1$, per-category state (h_k, T_k, c_k) evolves independently under thermodynamics, then the Lipscomb (2001) linear remapping redistributes ice across thickness categories, conservatively maintaining a well-resolved thickness distribution without the numerical diffusion of a simple rebinning.

10.4 Free-Drift Dynamics

When `dynamics="free_drift"`, ice velocity is diagnosed from ocean and atmospheric drag:

$$\mathbf{u}_i = C_{\text{ocean}} \mathbf{u}_{\text{ocean}} + \frac{C_{\text{atm}} \rho_{\text{air}}}{\rho_{\text{ice}}} \mathbf{u}_{\text{wind}}, \quad (10.6)$$

with $C_{\text{ocean}} = 5.5 \times 10^{-3}$, $C_{\text{atm}} = 1.3 \times 10^{-3}$.

10.5 Ice Tracer Transport

When `transport="advect"`, ice thickness, concentration, and temperature are advected by the ice velocity field. For multi-category ice, advection is applied independently to each category via `jax.vmap`.

Chapter 11

Tile-Based Coupler

11.1 Surface Flux Blending

Fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (11.1)$$

Here X is any surface flux (momentum, sensible heat, latent heat, or a radiative quantity) and $f_{\text{ocn}}, f_{\text{land}}, f_{\text{ice}}, f_{\text{lake}}$ are the fractional areas of each subgrid tile within the grid cell. Each tile carries its own surface temperature and roughness and is run through its own bulk-flux closure, then the fluxes are aggregated by area; the constraint $\sum f_i = 1$ guarantees that the blended flux conserves the grid-cell energy and momentum budgets.

11.2 Ocean Albedo

The ocean surface albedo uses the [Briegleb et al. \(1986\)](#) zenith-angle-dependent formula:

$$\alpha_{\text{ocn}} = \frac{0.026}{1.1 \mu_0^{1.7} + 0.065} + 0.15 (\mu_0 - 0.1)(\mu_0 - 0.5)(\mu_0 - 1.0), \quad (11.2)$$

where $\mu_0 = \cos \theta_z$ is the cosine of the solar zenith angle. Albedo rises sharply at low sun ($\mu_0 \rightarrow 0$), reproducing the strong glint of the ocean under grazing illumination that a constant albedo would miss.

11.3 Lake Model

A two-layer model with epilimnion (warm, mixed, default depth $h_{\text{epi}} = 5$ m) and hypolimnion (cold, deep, $h_{\text{hypo}} = 20$ m).

11.3.1 Energy Balance

The epilimnion temperature evolves under the surface energy budget:

$$\rho_w c_w h_{\text{epi}} \frac{dT_{\text{epi}}}{dt} = \text{SW}_{\text{net}} + \text{LW}_{\text{net}} - \text{SH} - \text{LH} - F_{\text{mix}}, \quad (11.3)$$

The mixed epilimnion of depth h_{epi} , density ρ_w , and specific heat c_w stores the net surface radiation and turbulent fluxes, less the downward mixing flux F_{mix} that ventilates heat into the hypolimnion. The mixing flux between the two layers is

$$F_{\text{mix}} = \rho_w c_w k_{\text{eff}} \frac{T_{\text{epi}} - T_{\text{hypo}}}{d_{\text{mid}}}, \quad (11.4)$$

with wind-enhanced effective diffusivity $k_{\text{eff}} = k_{\text{mix}} (1 + \alpha_{\text{wind}} |\mathbf{V}|)$.

11.3.2 Freezing

Lake temperature is clamped at $T_f = 273.15$ K, the freezing point of fresh water, below which further cooling is diverted into ice formation rather than temperature. Two surface bulk-flux options are provided: fixed exchange coefficients, or an iterative scheme based on Monin–Obukhov similarity theory (Monin and Obukhov, 1954), which makes the turbulent exchange coefficients depend on the surface-layer stability rather than holding them constant.

Chapter 12

Machine Learning Components

12.1 Spherical Fourier Neural Operator (SFNO)

The Spherical Fourier Neural Operator (Bonev et al., 2023) extends the Fourier neural operator (Li et al., 2021) from the periodic torus to the sphere by replacing the planar FFT with the spherical-harmonic transform, so the learned convolution kernel is rotationally consistent and free of the seam artifacts a flat-FFT operator would imprint at the poles. Because the spectral convolution acts globally in one layer, the operator captures the long-range teleconnections that a purely local convolution cannot, which is what makes it attractive as a learned dynamical core or hybrid physics emulator in the spirit of Kochkov et al. (2024).

12.1.1 Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. **N processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** output = input + decoder($\dots$).

12.1.2 Spectral Convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (12.1)$$

where $\hat{f}_{nm}^{\text{in}}$ and $\hat{f}_{nm}^{\text{out}}$ are the input and output spherical-harmonic coefficients at total wavenumber n and zonal wavenumber m , and $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair. Mixing channels independently at each (n, m) makes the operation a true spectral convolution: it is diagonal in the harmonic index, so the cost stays linear in the number of retained modes while the kernel still couples every grid point through the inverse transform.

The SFNO can serve as a learned dynamical core or physics emulator within the legoESM framework, sharing the same spectral transform infrastructure as the conventional spectral cores.

Chapter 13

Data Assimilation

13.1 Incremental 4D-Var

legoESM provides a fully implemented incremental 4D-Var data assimilation system (Courtier et al., 1994; Rabier et al., 2000), leveraging JAX’s automatic differentiation to compute exact adjoints without hand-coded tangent-linear or adjoint models. The incremental formulation of Courtier et al. (1994) linearises the analysis problem about a background trajectory and solves for an increment, which keeps the inner minimisation quadratic and tractable while the outer loop relinearises about the updated state; here the JAX adjoint supplies the exact gradient that the inner loop requires.

13.1.1 Control Vector

The control vector $\mathbf{x}$ comprises the prognostic atmospheric fields:

- Horizontal winds (u, v) ;
- Temperature T ;
- Specific humidity q ;
- Surface pressure p_s .

The control vector specification supports both grid-point and spectral representations, with configurable variable subsets for targeted assimilation.

13.1.2 Cost Function

The cost function minimises the sum of background and observation terms:

$$J(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x} - \mathbf{x}_b) + \frac{1}{2} \sum_i (\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i)^T \mathbf{R}_i^{-1}(\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i), \quad (13.1)$$

where $\mathbf{x}_b$ is the background state, $\mathbf{B}$ is the background error covariance, $\mathcal{M}_{t_0 \rightarrow t_i}$ is the forward model propagating the state from the analysis time t_0 to observation time t_i , $\mathcal{H}_i$ are observation operators, $\mathbf{y}_i$ are observations, and $\mathbf{R}_i$ are observation error covariances. The first term penalises departure from the prior weighted by its inverse covariance; the second penalises model–observation mismatch summed over the assimilation window, so the minimiser of J is the trajectory that best reconciles the background with all observations distributed in time (Rabier et al., 2000).

13.1.3 Background Error Covariance

Four $\mathbf{B}$ matrix families:

- **Diagonal:** $\mathbf{B} = \text{diag}(\sigma_b^2)$;
- **Diffusion:** Implicit diffusion-based correlation modelling;
- **Spectral:** Diagonal in spectral space (scale-dependent variances);
- **Hybrid:** Linear combination of flow-dependent ensemble covariance and static $\mathbf{B}$.

The hybrid family blends a static climatological $\mathbf{B}$ with a flow-dependent ensemble estimate; because a finite ensemble produces spurious long-range correlations, the ensemble term is tapered by a compactly supported Gaspari–Cohn correlation function (Gaspari and Cohn, 1999), which sets distant covariances exactly to zero while keeping the localised matrix positive definite.

13.1.4 Minimisation

L-BFGS and conjugate gradient solvers with optional $\mathbf{B}^{1/2}$ preconditioning for improved convergence. The gradient $\nabla_{\mathbf{x}} J$ is computed exactly via `jax.grad` through the full forward model trajectory.

13.1.5 Observation Operators

Three observation operator types:

- **Direct:** Identity mapping for in-situ observations on model grid points;
- **Interpolating:** Bilinear/trilinear interpolation from model grid to observation locations;
- **Composite:** Chained operators for derived quantities (e.g., satellite radiances via a forward radiative transfer step).

All operators are JAX-differentiable, so `jax.grad` propagates through observation-space transformations seamlessly.

13.1.6 Analysis–Forecast Cycling

Sequential cycling: the analysis from one assimilation window becomes the background for the next, with checkpoint/restart support.

13.1.7 Incremental Analysis Update

To reduce initialisation shocks, the analysis increment $\delta\mathbf{x} = \mathbf{x}_a - \mathbf{x}_b$ is applied gradually over an IAU window of N_{IAU} timesteps:

$$\mathbf{x}^{n+1} = \mathcal{M}(\mathbf{x}^n) + \frac{\delta\mathbf{x}}{N_{\text{IAU}}}, \quad n = 0, \dots, N_{\text{IAU}} - 1. \quad (13.2)$$

Here $\mathbf{x}_a$ and $\mathbf{x}_b$ are the analysis and background states and $\delta\mathbf{x}/N_{\text{IAU}}$ is the fraction of the increment injected each step. Spreading the correction over the window lets the model’s fast adjustment processes absorb it in balance, so the gravity-wave noise that an instantaneous insertion would excite is suppressed; this produces dynamically balanced analyses and is the default cycling mode.

Chapter 14

Conservation and Numerical Corrections

14.1 Atmospheric Mass Conservation

Because surface pressure measures the column-integrated mass, any residual imbalance left by the transport and physics updates appears as a drift in the global mean p_s . Dry-air mass is restored by a uniform adjustment to p_s :

$$\Delta\bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - M_{\text{target}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta\bar{p}. \quad (14.1)$$

Here A_i is the cell area, $p_{s,i}^{\text{new}}$ the uncorrected surface pressure, and M_{target} the area-integrated mass to be conserved; $\Delta\bar{p}$ is the area-weighted mean error subtracted uniformly from every column. A spatially constant offset removes the global drift without altering horizontal pressure gradients, so the geostrophic balance and wind field are left untouched.

Initial-mass anchoring. For long integrations, $M_{\text{target}} = M_0 = \sum_i A_i p_{s,i}^{\text{init}}$ prevents cumulative $O(\varepsilon)$ drift. All accumulations use float64.

Moisture conservation. Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$, where Q_{old} and Q_{new} are the global water masses before and after the update. Multiplicative rescaling preserves the sign and relative vertical structure of q , so positivity is maintained while the total water is exactly restored.

14.2 Ocean Conservation

The ocean’s free surface plays the role surface pressure plays in the atmosphere: the global-mean elevation η records the total volume, so a volume drift is removed by a uniform shift of η . Volume: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt: uniform additive correction per unit volume. All three grids (cubed-sphere, lat-lon, MPAS) compute corrections simultaneously from the original pre-fix state and upcast to accumulation precision before global reductions.

Note: These are uniform additive fixers that violate local conservation. The proper solution is flux-form tracer advection with barotropic–baroclinic flux reconciliation.

Chapter 15

Differentiable Training

Because every component is written in differentiable JAX, the gradient of a trajectory loss with respect to model parameters is available exactly through the compiled time-stepping kernel, with no hand-coded adjoint. This turns parameter estimation, neural-network coupling, and hybrid learned-physical modelling into a single optimisation problem. The `training/` module provides end-to-end differentiable training pipelines for three modes:

15.1 Physics Parameter Tuning

`TrainablePhysicsParams` wraps 8 physics parameters (convective relaxation time, mixing length, drag coefficients, etc.) as an Equinox module with sigmoid-based constraints mapping from unconstrained space to valid physical ranges. Gradients flow through `eqx.filter_value_and_grad` applied to the model's `build_segment_fn(...).raw` (non-JIT, non-donating variant for AD compatibility).

15.2 Neural GCM Training

Spectral neural GCM training couples a learned column model with the spectral dynamical core. The loss function combines area-weighted MSE, spectral loss (penalising errors at each total wavenumber), and per-variable MSE, imported from `ml/loss.py`. Optimisation uses `ml/training.create_optimizer` (warmup + cosine decay + gradient clipping).

15.3 SFNO-Dycore Coupling

The Spherical Fourier Neural Operator can be coupled to any dynamical core for hybrid learned-physical simulations. Training uses the same segment-based architecture, with `SegmentForcing` as an explicit argument (not closure-captured) to prevent JIT recompilation when forcing data changes.

15.4 ERA5 Data Ingestion

`era5_to_state.py` converts ERA5 reanalysis (lat-lon) to any model grid with a local Zarr cache for efficient repeated access. Vertical interpolation handles pressure-to-sigma/hybrid coordinate mapping.

Chapter 16

Parallel Runtime

16.1 Execution Modes

Mode	Ranks	Devices/Rank	Halo Backend	Reduction Backend
serial	1	1	local	local
multi_device	1	N	JAX SPMD	local
mpi	N	1	MPI	MPI
hybrid	N	M	MPI + JAX	MPI + JAX

The `ParallelRuntime.create()` entry point detects MPI rank/size, validates device counts, and routes to the correct halo/reduction backend.

16.2 Domain Decomposition

Cubed sphere. Valid device counts: 1, 2, 3, 6, and $6k^2$ for $k \geq 2$. Each face can be further tiled into sub-face patches. `initialize_distributed(global_n=N) → scatter_to_local()` distributes the global state to rank-local patches; `gather_to_global()` is used only for I/O.

Voronoi/MPAS. The unstructured cell graph is partitioned by a capability-aware `method="auto"` default: METIS k -way graph partitioning (via the optional `pymetis` dependency, `pip install -e ".[mesh]"`) when available, otherwise the dependency-free recursive coordinate bisection (RCB) geometric partitioner. Graph partitioning minimizes the edge cut, reducing both load imbalance and halo (communication) volume on irregular or variable-resolution meshes; on the near-uniform icosahedral mesh RCB is already well balanced, so the METIS advantage grows with rank count (the boundary/interior ratio rises). A third `method="sfc"` orders cells along a Hilbert space-filling curve and assigns balanced contiguous chunks — a dependency-free yet locality-preserving alternative. Independently, `reorder_voronoi_for_sharding` orders cells *within* each shard along the Hilbert curve so every contiguous `NamedSharding` chunk is spatially compact (cache and GPU memory locality). Each rank's local arrays use *owned-first* indexing (owned cells precede halo cells), so owned-cell loops are a contiguous slice. A two-layer halo supports biharmonic operators.

16.3 MPI Differentiability

All halo exchange `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Global reductions use AD-safe `allreduce(SUM)` with a custom VJP; `MAX`, `MIN`, `allgather`, and `bcast` are not differentiable and are used only in diagnostics. Conservation fixers (`fix_mass`, `zero_mean_tendency`) correctly flow gradients through global reductions.

The native 4D halo exchange path (`pad_halo_4d`, `pad_halo_vector_4d`) exchanges all vertical levels in one MPI message per neighbour, replacing the earlier per-level `vmap(pad_halo)` which issued n_{lev} separate messages.

16.4 Ensemble Parallelism

`jax.vmap` across ensemble members with `jax.lax.scan` over time steps. Support for perturbed initial conditions, ensemble mean/std/percentile diagnostics, and gradient checkpointing.

16.5 Multi-Device Collective Reductions

The 2026-05 GPU-scaling branch (#228) compresses per-RK3 step collectives across all dycore backends. On the cubed-sphere finite-volume path the four sequential allreduces (one per prognostic variable) collapse to a single fused reduction with shape (B, h, h, n_{var}) , cutting halo bandwidth by $\sim 36\text{--}144\times$ at high resolution. The spectral primitive-equation core fuses the FFT and Legendre transforms into a single pass and stacks per-tracer spherical-harmonic analyses; on MPAS, MPI `sendrecv`s drop $12 \rightarrow 6$ per substep and mass-fix payloads $3 \rightarrow 2$. Fifteen SPMD-correctness unit tests gate the changes and ensure SPMD-trace stability under JAX 0.10. Real-hardware GPU/MPI validation on the Levante cluster is described in `docs/REAL_HARDWARE_SCALING.md` and `docs/GPU_SCALING_BRANCH.md`.

Chapter 17

CMIP Infrastructure

17.1 CF/CMOR-Compliant Output

CF-1.8 NetCDF4 files with CMIP6 Data Reference Syntax (DRS) filenames: `<var>_<table>_<model>_<expt>_<va`
Four CMOR tables: Amon (33 atmospheric variables including clear-sky radiative fluxes, cloud condensate, and relative humidity), Lmon (6 land variables), Omon (2 ocean variables: SST and sea-ice fraction), and Aday (5 daily atmospheric variables) on a standard 19-level pressure grid.

17.2 Experiment Templates

Template	Period	Forcing	CO ₂ range
piControl	1850–∞	Fixed 1850	284.3 ppmv
historical	1850–2014	Transient	284→397 ppmv
ssp245	2015–2100	SSP2-4.5	397→600 ppmv
ssp585	2015–2100	SSP5-8.5	397→1135 ppmv
amip	1979–2014	Prescribed SST	Transient
1pctCO2	1–150	1%/yr increase	284→1277 ppmv

17.3 AMIP CMIP6 Forcing Deck

The CMIP6 AMIP protocol prescribes a six-channel external-forcing stack on top of monthly SST/SIC. legoESM consumes the canonical input4MIPs/MPI-M files directly through six `NamedTuple` configs in `src/legoesm/forcing/external.py`, all of which carry an explicit `start_year` so multi-year files are sampled on their actual calendar timeline rather than collapsed onto a 12-month cyclic axis.

Forcing channels.

Channel	Config (<code>external.py</code>)	Canonical CMIP6/MPI-M file
SST/SIC	AMIPForcingConfig	HadISST/COBE monthly (t, ϕ, λ)
GHG	GHGConfig	greenhouse_historical_plus.nc (annual)
Ozone	OzoneConfig	vmro3_input4MIPs_ozone_CMIP_UReading-CCMI-1-0_*
Solar	SolarConfig	swflux_14band_cmip6_1850-2299-v3.2.nc
Aerosol	AerosolConfig	Kinne aeropt_kinne_sw_b14_*_rast.nc
Volcanic	AerosolConfig.volcanic_*	bc_aeropt_cmip6_volc_lw_b16_sw_b14_*

Time-axis dispatch. For every channel, the loader inspects the file's CF `time:units` and `calendar` attributes via `_load_time_anchor` and maps the simulation day onto the file's absolute axis through `_simday_to_file_day`. Cyclic 12-month climatologies and multi-year transient series are dispatched automatically: ozone files with $n_t > 12$ records preserve interannual evolution; the volcanic file's 12-month layout per year is sampled at the active calendar month so that 1982 (El Chichón) and 1991 (Pinatubo) appear at the correct simulation dates.

GHG annual file. Each of $\{\text{CO}_2, \text{CH}_4, \text{N}_2\text{O}, \text{CFC-11}, \text{CFC-12}\}$ is shape $(n_t, 1, 1)$ with time units "year as %Y.%f". Linear interpolation in fractional years yields scalar concentrations passed to RRTMG as volume mixing ratios via `ghg_concentrations_to_vmr`. Unit scales (1e-6, 1e-9, 1e-12) match input4MIPs convention. GHG concentrations are consumed by RRTMG only — the gray-radiation pathway uses a constant CO_2 optical depth and does not respond to the prescribed time series.

Ozone. O_3 volume mixing ratio $X_{\text{O}_3}(t, \phi, p)$ is loaded with auto-detection of variable name (`vmro3`, `tro3`, `O3`, `ozone`) and unit (mol mol^{-1} vs. kg kg^{-1} auto-converted). The CMIP6 layout (t, p, ϕ, λ) is auto-swapped to (t, ϕ, p) after a zonal mean. Vertical interpolation in `_interp_vertical` handles both ascending and descending pressure axes (CMIP6 files ship $1000 \rightarrow 0.1$ hPa).

Solar TSI and 14-band SW. The MPI-M file provides daily $\text{TSI}(t)$ in W m^{-2} and $\text{SSI_frac}(t, n_{\text{band}}=14)$ summing to one. When `source="spectral_file"`, the 14 RRTMG-SW bands are expanded to 112 g -points by uniform fill within each band via `_expand_bands_to_gpoints`; the per- g -point fractions are passed into the SW solver to modulate the band-integrated insolation. Solar geometry (zenith angle, declination) is computed analytically and is not part of the prescribed forcing.

Aerosol and volcanic AOD. Background AOD is a zonal-mean monthly climatology $(\text{aod}_{550})(t, \phi)$. Multi-band Kinne files $(t, n_{\text{band}}, \phi, \lambda)$ are reduced to (t, ϕ) by zonal mean and band sum. The volcanic channel adds a second AOD field on top, with optional scaling `volcanic_scale`; both are clipped to $\text{AOD} \geq 0$ before reaching the SW radiation pathway.

Deck driver. `scripts/run_amip_cmip6_deck.py` pins a CMIP6 AMIP physics stack of RRTMG radiation, Sundqvist cloud fraction, Sundqvist large-scale condensation (the deck microphysics default — Kessler in the integrated AMIP path produces NaN winds at day ~ 2 and is tracked as a known issue), SBM convection, and Louis turbulence, and fixes `--ghg-forcing external`, `--ozone-forcing external`, `--solar-source spectral_file`, `--solar-tsi-var TSI`, `--solar-spectral-var SSI_frac`. When forcing files are absent the user must opt in with `--auto-generate` to invoke `scripts/generate_amip_forcing.py` for synthetic CMIP6-shape replacements; otherwise

the deck refuses to run. `scripts/validate_amip_run.py` reads the resulting CMOR output for post-run sanity.

Multi-grid dispatch coverage. `scripts/smoke_test_amip_all_grids.py` exercises the deck *dispatch path* on cubed-sphere (centred / FV3 / C-D), lat-lon (centred / FV3 / C-grid), Gaussian spectral, and Voronoi/MPAS. For run-time bound and stability the smoke test uses gray radiation with aerosol and volcanic disabled (`--radiation gray --no-aerosol --no-volcanic`); it confirms the (grid, discretisation, forcing-loader) machinery wires up but does *not* validate the full forcing stack on every grid. The Voronoi/MPAS branch additionally requires `--turbulence none` (the TKE bridge expects cell-centred winds, MPAS stores edge-normal winds) and `--dt 60` (the MPAS hydrostatic dycore is unstable at the default 600 s step despite a CFL diagnostic of 0.09). Spectral and MPAS run paths bypass `DiagnosticCollector`, so detailed zonal/profile diagnostics are unavailable on those grids; production science runs target cubed-sphere or lat-lon.

17.4 Restart and Reproducibility

Each restart file includes a `.meta.json` sidecar with SHA-256 state digest, config hash, platform identifier, JAX version, x64 mode, and git hash.

Chapter 18

Verification and Testing

The legoESM codebase comprises approximately 385 source modules ($\sim 102,000$ lines of code), validated by 3,489 tests across 216 test files.

18.1 Williamson Standard Test Cases

The shallow water dynamical cores are verified against the standard Williamson test suite (Williamson et al., 1992), the community benchmark set that isolates the distinct error modes of a horizontal core:

- **Case 1:** Advection of a cosine bell;
- **Case 2:** Global steady-state nonlinear geostrophic flow;
- **Case 5:** Zonal flow over an isolated mountain;
- **Case 6:** Rossby–Haurwitz wave.

Each case probes a different property: Case 1 has an exact solution so it measures pure advective accuracy and shape preservation; Case 2 is a steady geostrophic state whose departure norm exposes spurious grid imprinting; Case 5 and the Rossby–Haurwitz wave of Case 6 test the nonlinear response to topography and the maintenance of a propagating planetary wave. A comprehensive diagnostic suite (`williamson_diagnostic.py`) computes L_1 , L_2 , and L_∞ error norms against analytical solutions where available, and performs cross-discretisation comparison across all five grid families (cubed-sphere C-D, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi).

18.2 Spectral Dynamical Core Test Suite

The spectral transform dynamical cores (shallow water and primitive equation) are validated by a dedicated comprehensive test suite covering:

- **Conservation:** Total mass, energy, and enstrophy conservation to machine precision (shallow water) and within bounded drift (primitive equation with hyperdiffusion);
- **Stability:** Long-integration stability under the dry Held–Suarez forcing (Held and Suarez, 1994) — Newtonian relaxation to a prescribed radiative-equilibrium temperature plus boundary-layer Rayleigh drag — at T21/L40 and T42/L40 resolutions, which exercises the core’s climate-statistics fidelity without any physics parameterisation;
- **Spectral convergence:** Verification that error norms decrease with increasing truncation wavenumber;

- **Semi-implicit correctness:** Validation that the implicit gravity-wave solver matches reference solutions and permits CFL numbers > 1 for gravity-wave modes.

18.3 Cross-Discretisation Intercomparison

All five shallow water discretisations are compared on the same test problems, verifying consistent convergence rates and inter-scheme agreement in error norms. This ensures that the shared physics interface produces consistent results regardless of the dynamical core choice.

18.4 MPI Distributed Tests

Dedicated distributed test suites (`tests/distributed/`) verify:

- **Halo exchange correctness:** Scalar and vector 4D halo exchange on 2–6 ranks matches serial reference within round-off;
- **MPI differentiability:** `jax.grad` through halo-exchange and allreduce operations produces gradients matching single-rank reference (`test_mpi_differentiability.py`);
- **MPI driver integration:** Full model stepping under MPI produces results consistent with serial execution (`test_mpi_driver.py`).

Chapter 19

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth mean radius	6.371×10^6	m
R_d	Dry-air gas constant	287.05	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air specific heat (c_p)	1004.64	$\text{J kg}^{-1} \text{K}^{-1}$
c_v	Dry-air specific heat (c_v)	717.59	$\text{J kg}^{-1} \text{K}^{-1}$
κ	Poisson constant R_d/c_p	0.2857	—
R_v	Water-vapour gas constant	461.51	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pv}	Specific heat of water vapour	1846	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pw}	Specific heat of liquid water	4218	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pi}	Specific heat of ice	2106	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
L_s	Latent heat of sublimation	2.834×10^6	J kg^{-1}
L_f	Latent heat of fusion	3.337×10^5	J kg^{-1}
ε	R_d/R_v	0.622	—
T_{freeze}	Freezing point of water	273.15	K
ρ_w	Liquid water density	1000	kg m^{-3}
ρ_i	Ice density	917	kg m^{-3}
p_0	Reference pressure	10^5	Pa
S_0	Total solar irradiance	1361	W m^{-2}
σ_{SB}	Stefan–Boltzmann constant	5.670×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
ρ_0	Ocean reference density	1025	kg m^{-3}

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